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Formalized threshold reversible hiding in encrypted images in Post-Quantum settings

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Abstract
As advancements in computer vision emerge, the necessity for securing image processing techniques also arises. For this purpose, one of the promising methods is Reversible Data Hiding in Encrypted Images (RDHEI), which directly secures visual data while providing the option of embedding auxiliary information. However, RDHEI has not been formalized in the cryptographic context. The absence of such formalization leads to a lack of provable security guarantees and robust privacy comparisons between schemes. In this work, we address this gap by formalizing RDHEI, where we formally define the syntax, correctness properties, and security notions, including semantic security, payload privacy, separability, compactness, and robustness. Then, we construct Post-Quantum RDHEI (PQ-RDHEI) as an efficient scheme secure against quantum adversaries under the Module Learning With Errors assumption in the quantum random oracle model. Our design uses a Module Lattice Key Encapsulation Mechanism, pseudorandom functions, hash functions, and verifiable secret sharing as building blocks. The quantitative experiments show that on the standard grayscale test images, our PQ-RDHEI achieves embedding rates of 0.61-2.27 bits per pixel against 0.04-0.18 bits per pixel for the most efficient existing scheme, along with a 13.4× improvement in mean rate, and raises the entropy of the encrypted shares from 7.48 to 7.9902-7.9992 bits per pixel.

Keywords Multimedia security systems · Reversible data hiding · Image encryption · Post-quantum cryptography · Provable security · Steganography

1 Introduction

Reversible data hiding (RDH) is a promising technique as it can embed auxiliary information in the cover media and at the same time allow the content's exact recovery [1–3]. It is this very property that makes RDH ideal for various applications where data and original media preservation are the main concerns, such as medical imaging, military reconnaissance, and

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legal document archiving [4–6]. Conventional RDH techniques use the approaches of difference expansion [1], histogram shifting [2], and prediction error expansion [3, 7] to work on plaintext images. These methods try to find a trade-off between the capacity to embed and the distortion of the cover image.

The rise of cloud computing, along with the growing demand for privacy protection, has led to the need for processing multimedia signals in the encrypted domain [8, 9]; recent developments underscore this evolution, spanning lightweight Feistel-based encryption for IoT security [10]; generally, it is common for image owners' systems to encrypt their media before uploading it to cloud storage or processing it in the cloud. Nevertheless, the providers of these services may have to insert metadata into the picture for such purposes as storage management, content annotation, or access control, but they want to carry out this task without decrypting the image. This situation inspired the idea of reversible data hiding in encrypted images (RDHEI) that was first put forward by Puech et al. [11] and improved by Zhang [12] later.

RDHEI schemes come under two main categories: proper room imposition after the ciphering (VRAE) [12–14] and room reservation prior to ciphering (RRBE) [15–17]. In the case of VRAE methods, initially, the image is encrypted, and then the space for embedding is created in the encrypted domain. On the other hand, the methods of RRBE provide space for embedding in the plaintext domain prior to encryption and simultaneously exploit spatial correlation in the original image to get a higher capacity [15]. Puteaux and Puech [16] introduced a scheme for RRBE that is both fast and is built on the idea of most significant bit (MSB) prediction. This was later enhanced by means of Huffman coding [17] and local difference prediction [18] methods.

To make it more difficult for the attackers to obtain the whole message even if they had various parts of the encrypted image, the secret sharing technique was adopted by various RDHEI schemes in their designs [19–23]. These approaches result in the distribution of the encrypted image into many shares so that a threshold number of shares is required for the reconstruction, even if some are defective or not available. Wu et al. [19] were the first to use secret sharing in RDHEI with the difference expansion. Chen et al. [22] further allowed multiple data hiders to be supported by this technique. Recently, Saeidi and Mirzakuchaki [24] introduced an RDHEI scheme based on Learning With Errors [25] (LWE)-based secret sharing utilizing adaptive Huffman coding.

Despite these advances, a fundamental limitation persists as stronger privacy analysis is not possible when the scheme is not formalized, and any security result will remain informal [26]. The existing RDHEI schemes do not have the necessary rigorous cryptographic definitions for the primitive itself and its security properties. Claims about confidentiality, integrity, and reversibility cannot be verified using standard cryptographic analysis without formal syntax and security models [27]. So, despite advances in both steganography and steganalysis, existing RDHEI schemes lack formal cryptographic treatment, which relies instead on heuristic security arguments without rigorous adversarial models or provable guarantees [28, 29]. This situation not only stops meaningful security comparisons but also makes it impossible to use reduction-based proofs that would relate the security of constructions to well-studied computational assumptions.

Moreover, completely encrypted systems and algorithms that are based on classical complexity assumptions might ultimately become obsolete due to the emergence of quantum computers [30, 31]. Recently, the National Institute of Standards and Technology (NIST)

has laid down the groundwork for post-quantum cryptographic algorithms [32], which in turn makes it imperative to come up with RDHEI schemes that are robust against quantum adversaries.

The urgency of this requirement for multimedia in particular follows from the *harvest-now-decrypt-later* threat as encrypted images intercepted or archived today can be stored by an adversary and decrypted once a cryptographically relevant quantum computer becomes available, so any image whose confidentiality must outlive the arrival of such a machine has to be protected with post-quantum primitives *now*. This is precisely the regime that RDHEI targets. Medical imaging archives are subject to statutory retention periods that commonly reach twenty to thirty years, and cloud-based radiology workflows require the storage provider to attach patient identifiers, study metadata and audit trails to the encrypted volumes without ever decrypting them; the same holds for military and satellite reconnaissance imagery, for archival cultural-heritage and legal-evidence collections, and for imagery from critical infrastructure such as power-grid and pipeline inspection, whose sensitivity likewise persists for decades. In all of these settings the two requirements that motivate RDHEI, annotating ciphertext in place and recovering the original pixel data exactly, coincide with a confidentiality horizon that classical encryption can no longer be assumed to cover, which is what makes a post-quantum RDHEI construction a practical rather than a merely theoretical concern.

We present the first formalization of Reversible Data Hiding in Encrypted Images as a cryptographic primitive and introduce security properties for it. To be more precise, our contributions are outlined as follows: First, a formal syntax for RDHEI is provided, which consists of the algorithms $(\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Embed}, \mathsf{Extract}, \mathsf{Dec})$, each with exact input-output specifications. Subsequently, properties of correctness (embedding correctness, decryption correctness) along with security notions (semantic security, payload privacy, robustness, separability, compactness) are formulated using game-based definitions. In the next step, we propose PQ-RDHEI, a post-quantum encryption scheme, which is not only efficient but also secure against quantum attacks. It is built on ML-KEM [32], pseudorandom functions [33], hash functions [34], and verifiable secret sharing [35]. The following step is to show that PQ-RDHEI possesses all the security features specified in the definitions under the Module Learning With Errors (MLWE) assumption in the quantum random oracle model (QROM). Finally, we illustrate via experimental results that PQ-RDHEI provides greater embedding capacity along with encrypted share entropy than the existing best [24] techniques.

The remainder of this paper is organized as follows. Section 2 presents the computational assumptions and cryptographic building blocks. In Section 3, we then define the formal syntax and security properties for RDHEI. The subsequent part, Section 4, presents our PQ-RDHEI construction. We provide efficiency analysis and experimental comparison, followed by formal security proofs in Sections 5 and 6. Finally, Section 7 concludes with a discussion of future directions.

2 Preliminaries

We denote by $\lambda \in \mathbb{N}$ the security parameter. For $n \in \mathbb{N}$, we write $[n] := \{1, 2, \ldots, n\}$. The notation $\{0,1\}^\lambda$ denotes the set of λ-bit strings, and $\{0,1\}^*$ denotes the set of all finite-length binary strings. We write $x \leftarrow_\$ S$ for uniform random sampling from a set S, and $y \leftarrow \mathcal{A}(x)$

for the output of a deterministic or randomized algorithm $\mathcal{A}$ on input x. The symbol $\|$ denotes string concatenation, and $\oplus$ denotes bitwise XOR. For a string s, we write $|s|$ for its bit-length. A function $f : \mathbb{N} \to \mathbb{R}$ is *negligible*, written $f(\lambda) = \mathsf{negl}(\lambda)$, if for every polynomial $p(\cdot)$ there exists λ_0 such that $f(\lambda) < 1/p(\lambda)$ for all $\lambda > \lambda_0$. We write PPT for probabilistic polynomial-time and QPT for quantum polynomial-time. The quantum random oracle model (QROM) is the idealized model where all parties have quantum superposition access to a random oracle. IND-CCA denotes indistinguishability under adaptive chosen-ciphertext attack. For images, W and H denote width and height in pixels, respectively.

2.1 Computational assumption

Definition 1 (Module Learning With Errors Problem [25]). Let λ be a security parameter, n a power of two, q a prime modulus, k the module rank, and $R_q = \mathbb{Z}_q[X]/(X^n + 1)$ the ring of polynomials modulo $X^n + 1$ with coefficients in $\mathbb{Z}_q$. Let χ be an error distribution over R_q (typically a centered binomial distribution β_η parameterized by η). The *Module Learning With Errors* ($\mathsf{MLWE}_{n,k,q,\chi}$) problem is defined as follows. Given m samples of the form $(\mathrm{a}_i, b_i) \in R_q^k \times R_q$ where either:

- $\mathrm{a}_i \leftarrow R_q^k$ is uniformly random and $b_i = \langle \mathrm{a}_i, \mathrm{s} \rangle + e_i \in R_q$ for a fixed secret $\mathrm{s} \leftarrow \chi^k$ and error $e_i \leftarrow \chi$ (MLWE distribution), or
- $(\mathrm{a}_i, b_i) \leftarrow R_q^k \times R_q$ are uniformly random (uniform distribution),

distinguish between the two cases with non-negligible advantage. The *decisional* MLWE *assumption* states that for any PPT adversary $\mathcal{A}$:

$$|\Pr[\mathcal{A}(\{(\mathrm{a}_i, \langle \mathrm{a}_i, \mathrm{s} \rangle + e_i)\}_{i=1}^m) = 1] - \Pr[\mathcal{A}(\{(\mathrm{a}_i, b_i)\}_{i=1}^m) = 1]| = \mathsf{negl}(\lambda),$$

where $\mathrm{s} \leftarrow \chi^k$, $e_i \leftarrow \chi$, $\mathrm{a}_i \leftarrow R_q^k$, and $b_i \leftarrow R_q$ are uniformly random.

2.2 Building blocks

Definition 2 (Pseudorandom Function [33]). A pseudorandom function family is a deterministic algorithm $\mathsf{PRF} : \{0,1\}^\lambda \times \{0,1\}^* \to \{0,1\}^*$ such that for any PPT distinguisher $\mathcal{D}$: $\left|\Pr_{K \leftarrow \{0,1\}^\lambda}[\mathcal{D}^{\mathsf{PRF}(K,\cdot)}(1^\lambda) = 1] - \Pr_{f \leftarrow \mathsf{Func}}[\mathcal{D}^{f(\cdot)}(1^\lambda) = 1]\right| = \mathsf{negl}(\lambda)$, where Func denotes the set of all functions from $\{0,1\}^*$ to $\{0,1\}^*$. We instantiate PRF using SHAKE256 as an extendable output function (XOF), which can produce arbitrary-length outputs.

Definition 3 (Hash Function [34]). A cryptographic hash function $\mathsf{H} : \{0,1\}^* \to \{0,1\}^\lambda$ satisfies collision resistance if for any PPT adversary $\mathcal{A}$: $\Pr[(x, x') \leftarrow \mathcal{A}(1^\lambda) : x \neq x' \wedge \mathsf{H}(x) = \mathsf{H}(x')] = \mathsf{negl}(\lambda)$. We instantiate H using SHA3-256 , which provides post-quantum collision resistance.

Definition 4 (Module Lattice Key Encapsulation Mechanism [32]). The key encapsulation mechanism ML-KEM, based on MLWE problem, with parameters

$(n, k, q, \eta_1, \eta_2, d_u, d_v)$ consists of ML-KEM = (KeyGen, Encaps, Decaps), where $\mathsf{Compress}_q(x, d) = \lfloor (2^d/q) \cdot x \rceil \bmod 2^d$ and $\mathsf{Decompress}_q(x, d) = \lfloor (q/2^d) \cdot x \rceil$ perform lossy compression and decompression to d bits.

- $(pk, sk) \leftarrow \mathsf{KeyGen}(1^\lambda)$: On input the security parameter 1^λ, the key generation algorithm samples a seed $\rho \leftarrow \{0,1\}^{256}$, generates matrix $\hat{A} \in R_q^{k \times k}$ from ρ using XOF, samples $\mathrm{s}, \mathrm{e} \leftarrow \beta_\eta^k$ from centered binomial distribution β_η, computes $\mathrm{t} = \hat{A}\mathrm{s} + \mathrm{e} \in R_q^k$, and outputs public key $pk = (\mathrm{t}, \rho)$ and secret key $sk = \mathrm{s}$.
- $(c, K) \leftarrow \mathsf{Encaps}(pk)$: On input a public key pk, the encapsulation algorithm samples $m \leftarrow \{0,1\}^{256}$ uniformly, computes $(K', r) = G(\mathsf{H}(pk) \| m)$ where G is a key derivation function, samples $\mathrm{r}', \mathrm{e}_1, e_2$ from β_η using r as seed, computes $\mathrm{u} = \hat{A}^T \mathrm{r}' + \mathrm{e}_1 \in R_q^k$ and $v = \mathrm{t}^T \mathrm{r}' + e_2 + \lfloor q/2 \rceil \cdot \mathsf{Decode}(m) \in R_q$, compresses to $c = (\mathsf{Compress}_q(\mathrm{u}, d_u), \mathsf{Compress}_q(v, d_v))$, and outputs shared key $K = \mathsf{KDF}(K' \| \mathsf{H}(c))$ and ciphertext c.
- $K \leftarrow \mathsf{Decaps}(sk, c)$: On input a secret key sk and ciphertext c, the decapsulation algorithm parses $c = (\mathrm{u}', v')$, decompresses to $\mathrm{u} = \mathsf{Decompress}_q(\mathrm{u}', d_u)$ and $v = \mathsf{Decompress}_q(v', d_v)$, computes $m' = \mathsf{Encode}(\lfloor (v - \mathrm{s}^T \mathrm{u}) \cdot 2/q \rceil \bmod 2)$, recomputes $(K', r) = G(\mathsf{H}(pk) \| m')$, verifies encapsulation by recomputing c' with (pk, m', r), and outputs $K = \mathsf{KDF}(K' \| \mathsf{H}(c))$ if $c = c'$, otherwise outputs a pseudorandom key derived from a secret rejection value.

Definition 5 (Verifiable Secret Sharing [35]). The hash-based verifiable secret sharing scheme Π_{LA}, operating in the quantum random oracle model with threshold (t, n) where $n \geq 2t + 1$, consists of $_{\mathsf{LA}} = (\mathsf{Share}, \mathsf{VerifyShare}, \mathsf{Reconstruct})$.

- $(\{s_i, \gamma_i\}_{i=1}^n, \pi_{\mathsf{share}}) \leftarrow \mathsf{Share}(s, t, n)$: On input secret $s \in \mathbb{Z}_N$, threshold t, and number of parties n, the sharing algorithm samples random degree-t polynomials $f(X), r(X) \in \mathbb{Z}_N[X]_t$ with $f(0) = s$, computes shares $s_i = f(i)$ and randomizers $\gamma_i = r(i)$ for $i \in [1, n]$, generates commitments $c_i = \mathsf{H}(s_i \| \gamma_i)$, computes challenge $d = \mathsf{H}(c_1 \| \cdots \| c_n)$, sets $z(X) = r(X) + d \cdot f(X)$, and outputs shares $\{s_i, \gamma_i\}_{i=1}^n$ and proof $\pi_{\mathsf{share}} = (z(X), c_1, \ldots, c_n)$.
- $\{0,1\} \leftarrow \mathsf{VerifyShare}(i, s_i, \gamma_i, \pi_{\mathsf{share}})$: On input index i, share s_i, randomizer γ_i, and $\pi_{\mathsf{share}} = (z(X), c_1, \ldots, c_n)$, the verification algorithm checks if $\deg(z(X)) \leq t$, computes $d = \mathsf{H}(c_1 \| \cdots \| c_n)$, verifies $c_i = \mathsf{H}(s_i \| z(i) - d \cdot s_i)$, and outputs 1 if all checks pass, else 0.
- $s \leftarrow \mathsf{Reconstruct}(\{(i, s_i, \gamma_i)\}_{i \in S}, \pi_{\mathsf{share}})$: On input a set of verified share tuples $\{(i, s_i, \gamma_i)\}_{i \in S}$ where $|S| \geq t + 1$ and π_{share}, the reconstruction algorithm verifies each share using VerifyShare, computes Lagrange coefficients $\lambda_i^S = \prod_{j \in S, j \neq i} \frac{j}{j - i} \pmod N$, reconstructs $s = \sum_{i \in S} \lambda_i^S \cdot s_i \pmod N$, and outputs s.

In our PQ-RDHEI in the next sections, ML-KEM is for key encapsulation to establish shared encryption keys, relying on MLWE hardness with the content owner holding the public key and secret key for decryption; SHAKE256 as a pseudorandom function for block mask and payload masking generation, with keys derived from the KEM shared secret; SHA3 for key derivation, commitment, and tag generation, assuming collision resistance in QROM without trusted setup; secret sharing $_{\mathsf{LA}}$ for payload integrity verification, relying on binding

properties in QROM where dealers generate shares and any $t + 1$ parties reconstruct; Fisher-Yates permutation (Definition 14) with PRF-seeded deterministic shuffle for block position randomization, recoverable with the encryption key; and Huffman coding (Remark 4) with adaptive codebook for MSB compression to increase embedding capacity, which provides information-theoretic guarantees with deterministic codebooks from block statistics.

As for trust assumptions, the content owner holds (ek, dk). The data hider receives (hk, xk) and can embed/extract without accessing plaintext. Under the standard key model ($hk = xk$), both operations are available to the same party. The QROM assumption models hash functions as quantum-accessible random oracles, which reflects the post-quantum security goal.

Remark 1 (From MLWE to PQ-RDHEI). Because the manuscript refers both to the MLWE assumption and to the ML-KEM primitive, we make the dependency chain explicit. (i) $\mathsf{MLWE}_{n,k,q,\chi}$ (Definition 1) is a *computational assumption*: no QPT algorithm distinguishes MLWE samples from uniform ones with non-negligible advantage. (ii) ML-KEM (Definition 4) is a *primitive* whose IND-CCA security is proved from MLWE in the QROM by applying a Fujisaki–Okamoto transform to the underlying IND-CPA public-key encryption scheme [32]; the reduction is not tight, and the standardised parameter sets already absorb the loss. (iii) The *KEM-DEM composition* (Remark 5) turns the IND-CCA KEM together with a PRF-based stream cipher into a hybrid encryption scheme for images. (iv) PQ-RDHEI is built on top of that hybrid scheme. Consequently, our theorems are stated as reductions to the *security properties* of the building blocks, IND-CCA security of ML-KEM, PRF security of SHAKE256, collision resistance of SHA3-256, binding of Π_{LA}, and only the first of these is in turn implied by MLWE. The accurate phrasing is therefore that PQ-RDHEI is secure *provided* ML-KEM *is IND-CCA secure*, which holds under MLWE in the QROM; we use the shorthand "secure under MLWE" only for the composite statement obtained by chaining (i)–(iv).

3 Syntax for reversible hiding in encrypted images

We treat RDHEI as a cryptographic construction by specifying its syntax through a tuple of polynomial-time algorithms, defining correctness via probabilistic guarantees, and formulating security properties through adversarial games in the following.

Definition 6 (RDHEI). Let $\mathcal{M}$ be a message space, $\mathcal{D}$ a payload space, and $\kappa : \mathbb{N} \to \mathbb{N}$ a capacity function. A *reversible data hiding in encrypted domain* scheme is a tuple of PPT algorithms $\mathsf{RDHED} = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Embed}, \mathsf{Extract}, \mathsf{Dec})$ defined as follows:

- $\mathsf{KeyGen}(1^\lambda) \to (ek, dk, hk, xk)$: On input security parameter λ, outputs encryption key ek, decryption key dk, hiding key hk, and extraction key xk.
- $\mathsf{Enc}(ek, m) \to c$: On input ek and $m \in \mathcal{M}$, outputs ciphertext c.
- $\mathsf{Embed}(hk, c, d) \to \hat{c}$: On input hk, ciphertext c, and payload $d \in \mathcal{D}$ with $|d| \leq \kappa(|m|)$, outputs marked ciphertext $\hat{c}$.
- $\mathsf{Extract}(xk, \hat{c}) \to (c, d)$: On input xk and marked ciphertext $\hat{c}$, outputs ciphertext c and payload d, or $\perp$.

– $\mathsf{Dec}(dk, c) \to m$: On input dk and ciphertext c, outputs $m \in \mathcal{M}$ or $\perp$.

Remark 2 (Key Distribution Model). The keys (ek, dk) are held by the content owner; (hk, xk) are held by the data hider. A receiver may possess any subset. If ek is public, the scheme supports public-key encryption.

Remark 3 (Threat Model). The adversary is an honest-but-curious cloud storage or processing provider, or a network eavesdropper, modelled as PPT in the classical setting and QPT in the post-quantum setting. It observes the encryption key ek (public), any number of ciphertexts c and marked ciphertexts $\hat{c}$ together with their public tags, and, since ek is public, it has chosen-plaintext access by construction; chosen-ciphertext access is granted through the decapsulation oracle inherited from the IND-CCA security of ML-KEM, and the adversary is explicitly allowed to modify marked ciphertexts, which is the capability quantified by the robustness game of Definition 13. The data hider is untrusted with respect to the image content (it never receives dk and therefore learns nothing about the plaintext beyond its dimensions) but is trusted to run Embed honestly, since a malicious hider can always refuse to embed; collusion between the content owner and the receiver is permitted and harmless, as together they hold every key of an honestly generated instance. Compromise of hk costs exactly the confidentiality and integrity of the payload, the adversary can then unmask $\tilde{d}$, re-embed a payload of its choice and forge tag, but it does not affect image confidentiality, which rests on dk alone; the consequences of the constraint $hk = xk$ are analysed in Remark 6.

Definition 7 (Correctness). An RDHEI scheme satisfies *embedding correctness* if for all λ, $m \in \mathcal{M}$, $d \in \mathcal{D}$ with $|d| \le \kappa(|m|)$:

$$\Pr\left[\begin{array}{c}(ek, dk, hk, xk) \leftarrow \mathsf{KeyGen}(1^\lambda);\ c \leftarrow \mathsf{Enc}(ek, m);\ \hat{c} \leftarrow \mathsf{Embed}(hk, c, d)\\ (c', d') \leftarrow \mathsf{Extract}(xk, \hat{c})\ :\ c' = c \wedge d' = d\end{array}\right] = 1 - \mathsf{negl}(\lambda).$$

Definition 8 (Decryption Correctness). An RDHEI scheme satisfies *decryption correctness* if for all λ, $m \in \mathcal{M}$:

$$\Pr\left[(ek, dk, hk, xk) \leftarrow \mathsf{KeyGen}(1^\lambda); c \leftarrow \mathsf{Enc}(ek, m)\ :\ \mathsf{Dec}(dk, c) = m\right] = 1 - \mathsf{negl}(\lambda).$$

Definition 9 (Semantic Security). An RDHEI scheme satisfies *semantic security* if for any PPT adversary $\mathcal{A}$, the advantage $\mathsf{Adv}^{\mathsf{sem}}_{\mathcal{A}}(\lambda) = \left|\Pr[\mathsf{Expt}^{\mathsf{sem}}_{\mathcal{A},0}(\lambda) = 1] - \Pr[\mathsf{Expt}^{\mathsf{sem}}_{\mathcal{A},1}(\lambda) = 1]\right|$ is negligible, where $\mathsf{Expt}^{\mathsf{sem}}_{\mathcal{A},b}(\lambda)$ is defined as: $(ek, dk, hk, xk) \leftarrow \mathsf{KeyGen}(1^\lambda)$. Give (ek, hk, xk) to $\mathcal{A}$. $\mathcal{A}$ outputs (m_0, m_1) with $|m_0| = |m_1|$. Challenger computes $c^* \leftarrow \mathsf{Enc}(ek, m_b)$ and gives c^* to $\mathcal{A}$. $\mathcal{A}$ outputs bit b'. Output b'.

Definition 10 (Payload Privacy). An RDHEI scheme satisfies *payload privacy* if for any PPT adversary $\mathcal{A}$ we have $\mathsf{Adv}^{\mathsf{pay}}_{\mathcal{A}}(\lambda) = \left|\Pr[\mathsf{Expt}^{\mathsf{pay}}_{\mathcal{A},0}(\lambda) = 1] - \Pr[\mathsf{Expt}^{\mathsf{pay}}_{\mathcal{A},1}(\lambda) = 1]\right| = \mathsf{negl}(\lambda)$, where $\mathsf{Expt}^{\mathsf{pay}}_{\mathcal{A},b}(\lambda)$ is defined by the following procedure: $(ek, dk, hk, xk) \leftarrow \mathsf{KeyGen}(1^\lambda)$. Give $(ek,\ hk)$ to $\mathcal{A}$. $\mathcal{A}$ outputs (m, d_0, d_1) with $|d_0| = |d_1| \le \kappa(|m|)$. $c \leftarrow \mathsf{Enc}(ek, m)$; $\hat{c}^* \leftarrow \mathsf{Embed}(hk, c, d_b)$. Give $\hat{c}^*$ to $\mathcal{A}$. $\mathcal{A}$ outputs b'. Output b'.

Definition 11 (Separability). An RDHEI scheme is *separable* if: (I) $\mathsf{Extract}(xk, \cdot)$ can be computed without knowledge of dk (extraction-first), and (II) there exists $\mathsf{Dec}'(dk, \hat{c}) \to \hat{m}$ computable without xk, where $\hat{m}$ encodes both m and d in a recoverable manner (decryption-first).

Definition 12 (Compactness). An RDHEI scheme is *compact* if there exists a polynomial $p(\cdot)$ such that for all valid (c, d) we have $|\mathsf{Embed}(hk, c, d)| \leq |c| + p(\lambda, |d|)$. The scheme is *strongly compact* if $|\mathsf{Embed}(hk, c, d)| = |c|$.

Definition 13 (Robustness). An RDHEI scheme satisfies *embedding robustness* if for any PPT adversary $\mathcal{A}$:

$$\Pr\left[\begin{array}{c}(ek, dk, hk, xk) \leftarrow \mathsf{KeyGen}(1^\lambda);\ (c, d, \hat{c}^*) \leftarrow \mathcal{A}(ek, dk, hk, xk);\ (c', d') \leftarrow \mathsf{Extract}(xk, \hat{c}^*) \\ \hat{c}^* \neq \mathsf{Embed}(hk, c, d) \ \wedge\ c' = c \wedge d' = d\end{array}\right] = \mathsf{negl}(\lambda).$$

Prior RDHEI schemes operated under informal security assumptions. We contextualize our formalized definitions:

- Existing works [12, 16, 24] use XOR-based ciphers or chaotic maps, assuming computational indistinguishability without formal reduction. Our semantic security (Definition 9) provides a game-based formulation reducible to IND-CCA security of the underlying KEM.
- Prior schemes [19, 22] protect payloads via embedding location obscurity rather than cryptographic guarantees. Our payload privacy (Definition 10) ensures that adversaries knowing the embedding algorithm cannot distinguish between payloads.
- Most RDHEI constructions lack integrity verification. Our robustness definition (Definition 13) ensures adversarially-crafted ciphertexts cannot be misattributed to honest embeddings, protecting against tampering attacks.
- The separability notion (Definition 11) formalizes extraction-first and decryption-first capabilities that prior works [14, 21] describe informally, enabling detailed comparison of operational modes.

To substantiate the priority claim rather than assert it, Table 1 audits the representative RDHEI literature against the individual components of a cryptographic treatment. The distinction we draw is deliberately narrow: a property is credited only when the paper in question states it as a definition with quantified adversary, experiment and advantage function, not when it is argued informally or supported by statistical evidence such as histogram, entropy or correlation tests. Read this way, the audit shows that the individual *ingredients* are not new. Separability was isolated as a design goal by Zhang [14] and is standard in later work; reversibility is universally stated, though as an algorithmic property rather than as a probabilistic correctness condition; embedding capacity is always quantified; and the secret-sharing line [19–23] inherits an information-theoretic threshold guarantee for the *shares*, which is a genuine formal statement, albeit one about the sharing layer alone and not about the marked ciphertext. What is absent throughout is the surrounding apparatus: no prior work gives a syntax as a tuple of algorithms with typed inputs and outputs, none formulates confidentiality as a game with an explicit advantage function, none proves a reduction from

Table 1 Components of a cryptographic treatment that are formally defined in representative RDHEI literature. ✓: stated as a formal definition or proved; ◦: addressed informally, by construction, or supported only by statistical evidence; ✗: absent. "IT" denotes an information-theoretic guarantee for the secret-sharing layer only

Scheme	Family	Formal syntax	Game-based confidentiality	Reduction to assumption	Payload privacy	Robustness /integrity	Separa-bility	Reversi-bility	Post-quantum
Zhang [12]	VRAE	✗	✗	✗	✗	✗	✗	◦	✗
Zhang [14]	VRAE	✗	✗	✗	✗	✗	◦	◦	✗
Liao & Shu [13]	VRAE	✗	✗	✗	✗	✗	✗	◦	✗
Ma et al. [15]	RRBE	✗	✗	✗	✗	✗	◦	◦	✗
Puteaux & Puech [16]	RRBE / MSB	✗	✗	✗	✗	✗	◦	◦	✗
Yin et al. [17]	RRBE / Huffman	✗	✗	✗	✗	✗	◦	◦	✗
Mohammadi et al. [18]	RRBE	✗	✗	✗	✗	✗	◦	◦	✗
Wu et al. [19]	Secret sharing	✗	✗	IT	✗	✗	◦	◦	✗
Chen et al. [20]	Secret sharing	✗	✗	IT	✗	✗	◦	◦	✗
Qin et al. [21]	Secret sharing	✗	✗	IT	✗	✗	◦	◦	✗
Chen et al. [22]	Secret sharing	✗	✗	IT	✗	✗	◦	◦	✗
Hua et al. [23]	Secret sharing	✗	✗	IT	✗	✗	◦	◦	✗
Ke et al. [9]	Homomorphic	✗	◦	◦	✗	✗	◦	◦	✗
Saeidi & Mirzakuchaki [24]	LWE / sharing	✗	✗	◦	✗	✗	◦	◦	◦
This work (PQ-RDHEI)	KEM-DEM / MLWE	✓	✓	✓	✓	✓	✓	✓	✓

a stated computational assumption, none defines payload privacy as a property separate from image confidentiality, and none defines an integrity notion under which adversarially modified marked ciphertexts must be rejected. Our contribution is therefore the assembly, and specifically the four columns of Table 1 that are empty for every prior row: game-based semantic security with a reduction, payload privacy as an independent notion, embedding robustness, and a post-quantum assumption. We restrict the claim accordingly in the remainder of the paper: PQ-RDHEI is the first RDHEI construction with reduction-based proofs against quantum adversaries, and the definitional framework of this section is the first for RDHEI as a primitive, but the underlying image-domain techniques, MSB prediction, adaptive Huffman coding of the location map, block permutation, are inherited from [16, 17, 24] and are not claimed as novel.

4 Proposed formalized RDHEI scheme

We now present our post-quantum RDHEI construction, beginning with auxiliary definitions, followed by the main design and its instantiation from the building blocks introduced in Section 2.

Definition 14 (Fisher-Yates Permutation [36]). The deterministic Fisher-Yates algorithm $\mathsf{FisherYates}(\ell, K)$ generates a pseudorandom permutation $\sigma : [\ell] \to [\ell]$ as follows: initialize array $A = [1, 2, \ldots, \ell]$; for $i = \ell$ down to 2: compute $j = (\mathsf{PRF}(K, i) \mod i) + 1$, swap $A[i] \leftrightarrow A[j]$; output permutation σ where $\sigma(k) = A[k]$.

Definition 15 (Embedding Capacity). The capacity function $\kappa : \mathbb{N} \to \mathbb{N}$ for PQ-RDHED with parameters (n_b, β, T) on image of size $W \times H$ is defined as: $\kappa(W \cdot H) = \sum_{k \in \mathcal{E}} \left(\beta \cdot n_b^2 - \sum_{u=1}^{n_b^2} |\mathsf{cw}_{k,u}| \right) - |\mathsf{overhead}|$, where $\mathcal{E}$ is the set of embeddable blocks determined by threshold T, and $|\mathsf{overhead}|$ accounts for the length encoding, auxiliary data, and verification proof. The expected capacity satisfies $\kappa(W \cdot H) = \Theta(W \cdot H \cdot \beta / n_b^2)$ for typical natural images.

Remark 4 (Adaptive Huffman Codebook). For embeddable blocks with $\alpha \in \{1, 2, 3, 4\}$ distinct MSB types, the Huffman codebook is predefined as follows:

α	1	2	3	4
MSB Rank	1	1, 2	1, 2, 3	1, 2, 3, 4
Codeword	1	1, 0	1, 01, 00	1, 01, 001, 000

The encoded result for pixel u in block k is $(0)^{p_{k,u}} \| 1 \| \overline{\mathsf{code}}$, where $p_{k,u}$ counts prefix zeros allocated for embedding and $\overline{\mathsf{code}}$ is the reversed Huffman codeword.

We now present our main construction. Let $\mathcal{M} = \{0, 1, \ldots, 255\}^{W \times H}$ denote the message space of grayscale images with dimensions $W \times H$, and let $\mathcal{D} = \{0, 1\}^*$ denote the payload space. We define the block size parameter $n_b \in \mathbb{N}$ (typically $n_b = 4$ or $n_b = 8$), the MSB parameter $\beta \in \{3, 4, 5, 6\}$ denoting the number of most significant bits analyzed, and the embeddability threshold $T \in \{2, 3, 4\}$.

Remark 5 Our construction employs a hybrid KEM-DEM (Key Encapsulation Mechanism - Data Encapsulation Mechanism) paradigm for image encryption. The ML-KEM component provides post-quantum secure key establishment, outputting a shared secret K. The DEM component is instantiated as a PRF-based stream cipher operating in counter mode: for each block index k, the encryption key K_e (derived from K) seeds the PRF to generate an 8-bit block mask $\mu_k = \mathsf{PRF}(K_e, k)[0:8]$, which is XORed with each pixel in block k. This design ensures that pixels within the same block share a common mask, thereby preserving intra-block MSB correlation structure essential for identifying embeddable regions during the data hiding phase.

To specify concrete parameters for PQ-RDHEI instantiation and analyze security levels, we instantiate ML-KEM at security level 3, providing approximately 192-bit classical and 128-bit quantum security [32] with parameters $n = 256, k = 3, q = 3329, \eta_1 = \eta_2 = 2, d_u = 10$, $d_v = 4$, yielding key and ciphertext sizes $|pk| = 1184$ bytes, $|sk| = 2400$ bytes, $|c_{\mathsf{kem}}| = 1088$ bytes, and $|K| = 32$ bytes. We set the parameter $\lambda = 256$ bits, and SHA3 provides 256-bit classical and 128-bit quantum collision resistance under Grover's algorithm [37]. SHAKE256 as an XOF provides PRF security with adaptable output length. Image processing parameters are: block size $n_b \in \{2, 4, 8\}$, MSB bits per pixel $\beta \in \{3, 4, 5, 6\}$, and embeddability threshold $T \in \{2, 3, 4\}$. For high-capacity embedding, we recommend $(n_b, \beta, T) = (4, 5, 4)$; for higher fidelity, $(n_b, \beta, T) = (2, 4, 2)$ provides more granular control. Under these parameters, the security advantages satisfy $\mathsf{Adv}^{\mathsf{sem}}_{\mathcal{A}}(\lambda) \leq 2^{-128} + \ell \cdot 2^{-256} \approx 2^{-128}$ and $\mathsf{Adv}^{\mathsf{robust}}_{\mathcal{A}}(\lambda) \leq 2^{-128} + 2^{-128} + q_H^2 \cdot 2^{-256} \approx 2^{-127}$ where $\ell \leq 2^{20}$ is the maximum number of blocks (supporting images up to 4096×4096 with $n_b = 4$) and $q_H \leq 2^{64}$ is the maximum number of hash queries, which ensures security margins well above the 100-bit threshold for long-term security.

Definition 16 (Post-Quantum RDHEI). The post-quantum reversible data hiding in encrypted domain scheme PQ-RDHED, instantiated from ML-KEM, PRF, H, and Π_{LA} with parameters (λ, n_b, β, T), consists of the tuple PQ-RDHEI = (KeyGen, Enc, Embed, Extract, Dec) as formally specified in Fig. 1. The message space is $\mathcal{M} = \{0, 1, \ldots, 255\}^{W \times H}$ for gray-scale images of dimensions $W \times H$, the payload space is $\mathcal{D} = \{0, 1\}^*$, and the capacity function κ is defined per Definition 15.

The scheme operates through a key generation, where ML-KEM.KeyGen establishes the asymmetric encryption infrastructure (pk, sk) while independent λ-bit seeds K_h and K_x are sampled for the hiding and extraction keys respectively. During encryption, the input image m undergoes spatial partitioning via $\mathsf{Partition}(\cdot, n_b)$, which segments the pixel array into $\ell = \lceil W/n_b \rceil \cdot \lceil H/n_b \rceil$ non-overlapping blocks of size $n_b \times n_b$. The KEM-DEM paradigm then proceeds: ML-KEM.Encaps produces encapsulation c_{kem} and shared secret K, from which domain-separated keys K_e and K_π are derived for block-wise stream cipher encryption and Fisher-Yates permutation, respectively. Each block receives a common 8-bit mask $\mu_k = \mathsf{PRF}(K_e, k)[0:8]$, preserving intra-block MSB correlation essential for subsequent embedding.

The embedding phase initiates with MSB analysis: for each encrypted block, the procedure extracts the β most significant bits of each pixel and counts distinct values via the set cardinality α_k. Blocks satisfying $\alpha_k \leq T$ are deemed embeddable and recorded in the loca-

PQ-RDHEI.KeyGen(1^λ)

$(pk, sk) \leftarrow$ ML-KEM.KeyGen(1^λ)
$K_h \leftarrow\$ \{0,1\}^\lambda$
$K_x \leftarrow K_h$
$ek \leftarrow pk$; $dk \leftarrow sk$
$hk \leftarrow K_h$; $xk \leftarrow K_x$
return (ek, dk, hk, xk)

PQ-RDHEI.Enc(ek, m)

$\{p_{i,j}\}_{i\in[W],j\in[H]} \leftarrow m$
$\ell \leftarrow \lceil W/n_b \rceil \cdot \lceil H/n_b \rceil$
$\{B_k\}_{k=1}^{\ell} \leftarrow$ Partition($\{p_{i,j}\}, n_b$)
$(c_{\mathsf{kem}}, K) \leftarrow$ ML-KEM.Encaps(ek)
$K_e \leftarrow \mathsf{H}(K\|\mathsf{0x00})$; $K_\pi \leftarrow \mathsf{H}(K\|\mathsf{0x01})$
for $k \in [\ell]$ **do**
  $\mu_k \leftarrow \mathsf{PRF}(K_e, k)[0:8]$
  for $u \in [n_b^2]$ **do**
    $e_{k,u} \leftarrow p_{k,u} \oplus \mu_k$
  $E_k \leftarrow \{e_{k,u}\}_{u\in[n_b^2]}$
$\sigma \leftarrow$ FisherYates(ℓ, K_π)
for $k \in [\ell]$ **do** $E'_{\sigma(k)} \leftarrow E_k$
$c \leftarrow (c_{\mathsf{kem}}, \{E'_k\}_{k=1}^{\ell})$
return c

PQ-RDHEI.Extract($xk, \hat{c}$)

$(c_{\mathsf{kem}}, \{\hat{E}_k\}_{k=1}^{\ell}, \mathsf{tag}) \leftarrow \hat{c}$
$K_{\mathsf{pos}} \leftarrow \mathsf{H}(xk\|\mathsf{0x02})$; $K_{\mathsf{mask}} \leftarrow \mathsf{H}(xk\|\mathsf{0x03})$
$\mathcal{E}' \leftarrow \emptyset$; $\mathsf{stream}' \leftarrow \varepsilon$
for $k \in [\ell]$ **do**
  if DetectPattern($\hat{E}_k, \beta$) $= 1$ **then** $\mathcal{E}' \leftarrow \mathcal{E}' \cup \{k\}$
for $k \in \mathcal{E}'$ **do**
  for $u \in [n_b^2]$ **do**
    $(p_{k,u}, \mathsf{code}_{k,u}) \leftarrow \mathsf{ParseMSB}(\hat{E}_k[u], \beta)$
    $\mathsf{stream}' \leftarrow \mathsf{stream}'\|(0)^{p_{k,u}}$
    $\mathsf{msb}_{k,u} \leftarrow \mathsf{Huffman}^{-1}(\overline{\mathsf{code}_{k,u}})$
$(|d|, \tilde{d}, \mathsf{aux}', \pi'_{\mathsf{share}}) \leftarrow \mathsf{Parse}(\mathsf{stream}')$
$b \leftarrow \Pi_{\mathsf{LA}}.\mathsf{VerifyShare}(1, s'_1, \gamma'_1, \pi'_{\mathsf{share}})$
if $b = 0$ **then return** $\perp$
$d \leftarrow \tilde{d} \oplus \mathsf{PRF}(K_{\mathsf{mask}}, \mathsf{0x00})[0:|d|]$
$h \leftarrow \Pi_{\mathsf{LA}}.\mathsf{Reconstruct}(\{(1, s'_1, \gamma'_1)\}, \pi'_{\mathsf{share}})$
if $h \neq \mathsf{H}(\mathsf{aux}'\|d)$ **then return** $\perp$
$(L', \{(\alpha_k, \{\mathsf{msb}_k^{(j)}\}_j)\}_k) \leftarrow \mathsf{aux}'$
for $k : L'[k] = 1$ **do**
  $E'_k \leftarrow \mathsf{RestoreMSB}(\hat{E}_k, \{\mathsf{msb}_{k,u}\}_u, \beta)$
for $k : L'[k] = 0$ **do** $E'_k \leftarrow \hat{E}_k$
if $\mathsf{tag} \neq \mathsf{H}(xk\|c_{\mathsf{kem}}\|\mathsf{H}(\mathsf{stream}'))$ **then return** $\perp$
$c \leftarrow (c_{\mathsf{kem}}, \{E'_k\}_{k=1}^{\ell})$
return (c, d)

PQ-RDHEI.Embed(hk, c, d)

$(c_{\mathsf{kem}}, \{E'_k\}_{k=1}^{\ell}) \leftarrow c$
$K_{\mathsf{pos}} \leftarrow \mathsf{H}(hk\|\mathsf{0x02})$; $K_{\mathsf{mask}} \leftarrow \mathsf{H}(hk\|\mathsf{0x03})$
$L \leftarrow \mathbf{0}^{\ell}$; $\mathcal{E} \leftarrow \emptyset$
for $k \in [\ell]$ **do**
  for $u \in [n_b^2]$ **do** $\mathsf{msb}_{k,u} \leftarrow \lfloor e_{k,u}/2^{8-\beta} \rfloor$
  $\alpha_k \leftarrow |\{\mathsf{msb}_{k,u} : u \in [n_b^2]\}|$
  if $\alpha_k \leq T$ **then** $\mathcal{E} \leftarrow \mathcal{E} \cup \{k\}$; $L[k] \leftarrow 1$
for $k \in \mathcal{E}$ **do**
  $(\mathsf{msb}_k^{(1)}, \ldots, \mathsf{msb}_k^{(\alpha_k)}) \leftarrow \mathsf{SortByFreq}(\{\mathsf{msb}_{k,u}\}_u)$
  $\mathsf{code}_k \leftarrow \mathsf{Huffman}(\alpha_k)$
  for $u \in [n_b^2]$ **do**
    $\mathsf{cw}_{k,u} \leftarrow (0)^{p_{k,u}}\|1\|\overline{\mathsf{code}_k(\mathsf{msb}_{k,u})}$
  $\mathsf{cap}_k \leftarrow \beta \cdot n_b^2 - \sum_{u=1}^{n_b^2} |\mathsf{cw}_{k,u}|$
$\mathsf{aux} \leftarrow L\|\|\mathcal{E}\|\|\{(\alpha_k, \{\mathsf{msb}_k^{(j)}\}_{j=1}^{\alpha_k})\}_{k\in\mathcal{E}}$
$(\{s_i, \gamma_i\}_{i=1}^2, \pi_{\mathsf{share}}) \leftarrow \Pi_{\mathsf{LA}}.\mathsf{Share}(\mathsf{H}(\mathsf{aux}\|d), 1, 2)$
$\tilde{d} \leftarrow d \oplus \mathsf{PRF}(K_{\mathsf{mask}}, \mathsf{0x00})[0:|d|]$
$\mathsf{stream} \leftarrow |d|_{32}\|\tilde{d}\|\mathsf{aux}\|\pi_{\mathsf{share}}$
$\mathsf{ptr} \leftarrow 0$
for $k \in \mathcal{E}$ **do**
  $\mathsf{emb}_k \leftarrow \mathsf{stream}[\mathsf{ptr} : \mathsf{ptr} + \mathsf{cap}_k]$
  $\mathsf{ptr} \leftarrow \mathsf{ptr} + \mathsf{cap}_k$
  $\hat{E}_k \leftarrow \mathsf{WriteEncoded}(E'_k, \{\mathsf{cw}_{k,u}\}_u, \mathsf{emb}_k, \beta)$
for $k \in [\ell] \setminus \mathcal{E}$ **do** $\hat{E}_k \leftarrow E'_k$
$\mathsf{tag} \leftarrow \mathsf{H}(hk\|c_{\mathsf{kem}}\|\mathsf{H}(\mathsf{stream}))$
$\hat{c} \leftarrow (c_{\mathsf{kem}}, \{\hat{E}_k\}_{k=1}^{\ell}, \mathsf{tag})$
return $\hat{c}$

PQ-RDHEI.Dec(dk, c)

$(c_{\mathsf{kem}}, \{E'_k\}_{k=1}^{\ell}) \leftarrow c$
$K \leftarrow$ ML-KEM.Decaps(dk, c_{kem})
$K_e \leftarrow \mathsf{H}(K\|\mathsf{0x00})$; $K_\pi \leftarrow \mathsf{H}(K\|\mathsf{0x01})$
$\sigma \leftarrow$ FisherYates(ℓ, K_π); $\sigma^{-1} \leftarrow \mathsf{Invert}(\sigma)$
for $k \in [\ell]$ **do** $E_k \leftarrow E'_{\sigma^{-1}(k)}$
for $k \in [\ell]$ **do**
  $\mu_k \leftarrow \mathsf{PRF}(K_e, k)[0:8]$
  for $u \in [n_b^2]$ **do**
    $p_{k,u} \leftarrow e_{k,u} \oplus \mu_k$
  $B_k \leftarrow \{p_{k,u}\}_{u\in[n_b^2]}$
$m \leftarrow \mathsf{Reconstruct}(\{B_k\}_{k=1}^{\ell}, W, H, n_b)$
return m

Fig. 1 Our Post-Quantum Reversible Data Hiding in Encrypted Image PQ-RDHEI = (KeyGen, Enc, Embed, Extract, Dec) based upon the standard building blocks Key Encapsulation ML-KEM=(KeyGen,Encaps,Decaps), Pseudorandom Function PRF, Hash Function H, and Verifiable Secret Sharing Π_{LA}=(Share,VerifyShare,Reconstruct) with parameters (n_b,β,T)

tion map L. The function SortByFreq orders the α_k distinct MSB values by occurrence frequency, enabling optimal Huffman code assignment per Remark 4. The auxiliary function $\mathsf{WriteEncoded}(E'_k, \{\mathsf{cw}_{k,u}\}_u, \mathsf{emb}_k, \beta)$ then performs an interleaving operation: it replaces each pixel's β-MSB with the concatenation of embedding bits (placed in prefix zero positions), a delimiter bit, and the reversed Huffman codeword. The complete embedding stream comprising length-prefixed masked payload $\tilde{d}$, auxiliary metadata aux, and VSS share π_{share}, is distributed across embeddable blocks in lexicographic order, culminating in a commitment tag tag binding the hiding key to the embedded content.

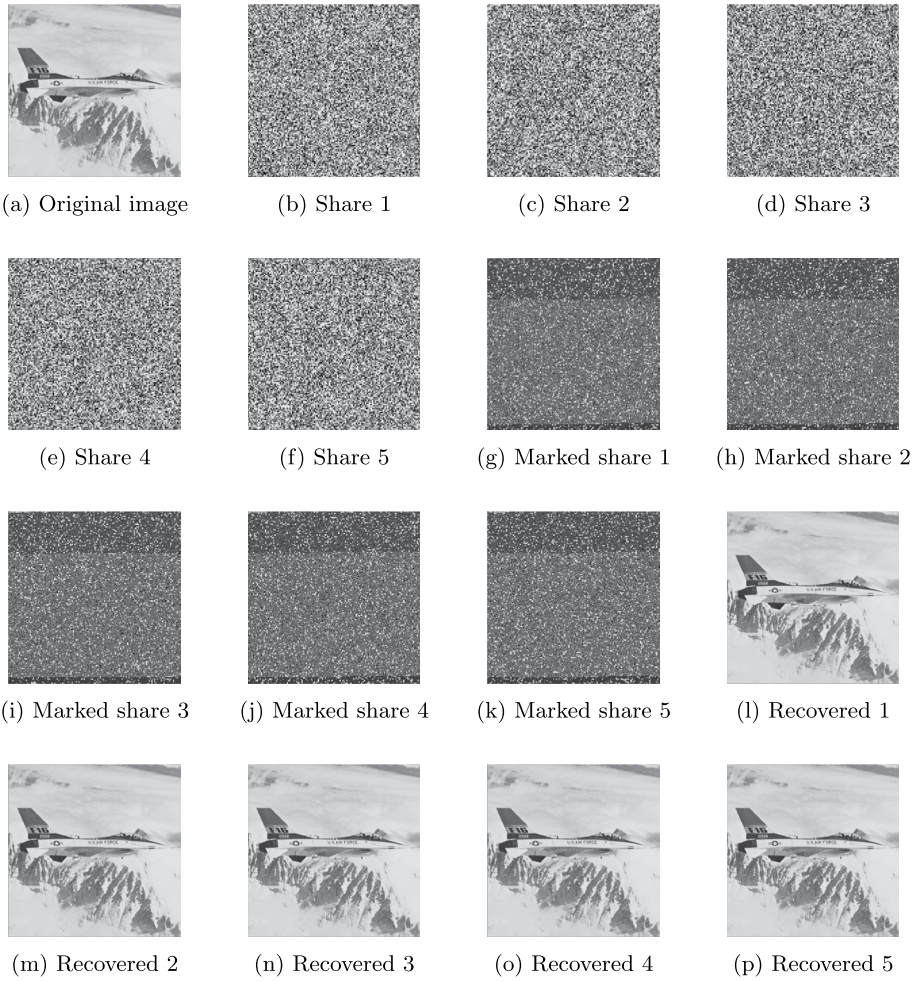
Extraction reverses this process through pattern detection: $\mathsf{DetectPattern}(\hat{E}_k, \beta)$ identifies embeddable blocks by scanning for the characteristic prefix-zero-delimiter structure in MSB positions. The function $\mathsf{ParseMSB}(\hat{E}_k[u], \beta)$ decomposes each pixel's upper bits into embedding prefix length $p_{k,u}$ and encoded codeword, while $\mathsf{Huffman}^{-1}$ recovers original MSB values via reverse Huffman decoding. After stream reassembly and VSS verification, $\mathsf{RestoreMSB}(\hat{E}_k, \{\mathsf{msb}_{k,u}\}_u, \beta)$ reconstructs the original encrypted blocks by reinstating decoded MSB values into their canonical positions. Decryption then proceeds conventionally: KEM decapsulation recovers K, inverse permutation σ^{-1} (computed via $\mathsf{Invert}(\sigma)$) restores block ordering, XOR with regenerated masks yields plaintext blocks, and finally $\mathsf{Reconstruct}(\{B_k\}_{k=1}^{\ell}, W, H, n_b)$ reassembles the original image from its constituent blocks (Fig. 2).

For images where dimensions are not divisible by n_b, boundary blocks are zero-padded with original dimensions stored in aux for exact recovery. The Extract algorithm returns $\perp$ upon tag verification failure, VSS verification failure, or malformed stream parsing, without leaking failure-mode information.

5 PQ-RDHED comparison and efficiency analysis

We begin with a broad comparison against representative schemes from each RDHEI family, and then narrow the quantitative evaluation to the most recent and most efficient of them. Table 2 places PQ-RDHEI against classical VRAE and RRBE constructions, the secret-sharing line, the homomorphic-encryption line and the recent LWE-based scheme of Saeidi & Mirzakuchaki [24], along the axes requested for a security-oriented comparison: embedding capacity and rate, security model, encryption primitive, post-quantum status, asymptotic complexity, runtime and memory, payload privacy, robustness and reversibility. Two observations follow. First, capacity and security are largely independent axes in this literature: the high-capacity RRBE schemes [17, 18] reach rates comparable to ours while offering no formal confidentiality guarantee at all, whereas the secret-sharing schemes buy an information-theoretic guarantee for the shares at a substantial capacity and bandwidth cost. Second, PQ-RDHEI is the only entry that is simultaneously in the high-capacity regime and equipped with reduction-based security, and the only one whose confidentiality survives a quantum adversary, the LWE-based scheme of [24] uses a lattice problem but derives no reduction from it, so its post-quantum status is heuristic.

A caveat on the runtime and memory columns: the figures reported in the cited papers were obtained on different processors, in different languages and on different image sets, so they are not directly comparable and we give only an order-of-magnitude classification. Only the row for PQ-RDHEI is measured here, under the conditions stated in Sec-



(a) Original image (b) Share 1 (c) Share 2 (d) Share 3

(e) Share 4 (f) Share 5 (g) Marked share 1 (h) Marked share 2

(i) Marked share 3 (j) Marked share 4 (k) Marked share 5 (l) Recovered 1

(m) Recovered 2 (n) Recovered 3 (o) Recovered 4 (p) Recovered 5

Fig. 2 Demonstration of our proposed Post-quantum RDHEI technique using the Jet sample image. (**a**) The original cover image. (**b**–**f**) Five encrypted shares generated via ML-KEM-based encryption, each exhibiting noise-like appearance with no discernible visual information. (**g**–threshold for long-term security.) Marked encrypted shares after data embedding, maintaining indistinguishability from unmodified encrypted shares. (l–p) Recovered images obtained by decrypting any four of the five marked shares, demonstrating perfect reconstruction with PSNR = $+\infty$ and SSIM = 1.0, confirming lossless reversibility of the scheme

tion 5.1. For the same reason, the detailed numerical comparison that follows is confined to a single reference method. It has been shown that Saeidi & Mirzakuchaki's technique is the most efficient RDHEI [24], so if our method is more efficient than Saeidi & Mirzakuchaki's technique, then our work is the most efficient RDHEI; thus, in the following, we focus on comparing our technique with Saeidi & Mirzakuchaki's method. We set up our experiments with parameters and specifications of Saeidi & Mirzakuchaki to facilitate an accurate comparison.

Table 2 Comparison of PQ-RDHEI with representative RDHEI schemes. Embedding rates are the values reported by the cited works for standard 512×512 grayscale test images and are indicative only, as test sets differ. Runtime and memory are order-of-magnitude classifications, since the cited measurements were taken on different platforms; only the PQ-RDHEI row is measured in this work (Section 5.1). "n" denotes the number of pixels and "s" the number of shares

Scheme	Family	Rate (bpp)	Security model	Encryption primitive	Post-quantum	Complexity	Run-time	Memory	Payload privacy	Robustness
Zhang [12]	VRAE	≈ 0.004	heuristic	stream cipher	✗	$O(n)$	low	$O(n)$	✗	✗
Zhang [14]	VRAE, separable	≈ 0.03	heuristic	stream cipher	✗	$O(n)$	low	$O(n)$	✗	✗
Ma et al. [15]	RRBE	≈ 0.5	heuristic	stream cipher	✗	$O(n)$	low	$O(n)$	✗	✗
Puteaux & Puech [16]	RRBE, MSB pred.	≈ 1.0	heuristic	stream cipher	✗	$O(n)$	low	$O(n)$	✗	✗
Yin et al. [17]	RRBE, Huffman	≈ 2.5	heuristic	stream cipher	✗	$O(n)$	low	$O(n)$	✗	✗
Mohammadi et al. [18]	RRBE, local pred.	≈ 2.6	heuristic	stream cipher	✗	$O(n)$	low	$O(n)$	✗	✗
Wu et al. [19]	Secret sharing	≈ 0.5	IT (shares)	(t, s) sharing	partial	$O(sn)$	mod.	$O(sn)$	✗	✗
Chen et al. [22]	Secret sharing	≈ 1.0	IT (shares)	(t, s) sharing	partial	$O(sn)$	mod.	$O(sn)$	✗	✗
Hua et al. [23]	Secret sharing, CFSS	≈ 2.4	IT (shares)	(t, s) sharing	partial	$O(sn)$	mod.	$O(sn)$	✗	✗
Qin et al. [21]	Secret sharing	≈ 1.2	IT (shares)	GF(p)/GF(2^8)	partial	$O(sn)$	mod.	$O(sn)$	✗	✗
Ke et al. [9]	Homomorphic	≈ 0.03	IND-CPA (FHE)	FHE	✓	$O(n \log n)$	high	$O(n)$	✗	✗
Saeidi & Mirzakuchaki [24]	LWE, sharing	0.04–0.18	heuristic	LWE sharing	heuristic	$O(sn)$	mod.	$O(sn)$	✗	✗
This work (PQ-RDHEI)	KEM-DEM, MLWE	0.61–2.27	IND-CCA + PRF, QROM	ML-KEM + PRF	✓	$O(n)$	91.7 ms (512^2)	$O(n)$	✓	✓

Table 3 Embedding capacity comparison ($\beta = 5$, $n = 8$, $T = 4$ for Saeidi & Mirzakuchaki [24]; $\beta = 5$, $n_b = 4$, $T = 4$ for our PQ-RDHEI)

	Saeidi & Mirzakuchaki [24]			Our PQ-RDHEI		
Sample Image	Entire Cap. (bits)	Pure Cap. (bits)	Rate (bpp)	Entire Cap. (bits)	Pure Cap. (bits)	Rate (bpp)
Airplane	489962	33786	0.1289	713955	595728	2.2725
Baboon	490680	33896	0.1293	160190	160190	0.6111
Barbara	517601	35899	0.1369	588832	588832	1.4198
Boat	262144	11263	0.0430	424278	424278	1.6185
Cameraman	726708	46269	0.1765	121155	121155	1.8487
Jet	664582	45652	0.1741	547026	531712	2.0283
Lake	489669	33791	0.1289	407241	407241	1.5535
Peppers	524288	23551	0.0898	573155	573155	2.1864

Table 4 PSNR and SSIM of recovered images, comparison with Saeidi & Mirzakuchaki [24]

	Saeidi & Mirzakuchaki [24]		Our PQ-RDHEI	
Sample Image	PSNR (dB)	SSIM	PSNR (dB)	SSIM
Airplane	$+\infty$	1.000000	$+\infty$	1.000000
Baboon	$+\infty$	1.000000	$+\infty$	1.000000
Barbara	$+\infty$	1.000000	$+\infty$	1.000000
Boat	$+\infty$	1.000000	$+\infty$	1.000000
Cameraman	$+\infty$	1.000000	$+\infty$	1.000000
Jet	$+\infty$	1.000000	$+\infty$	1.000000
Lake	$+\infty$	1.000000	$+\infty$	1.000000
Peppers	$+\infty$	1.000000	$+\infty$	1.000000

From an embedding capacity perspective, PQ-RDHEI has better performance across all test images. At the same time, both methods achieve equivalent perfect reversibility. However, PQ-RDHEI additionally exhibits higher encrypted share entropy, indicating stronger encryption properties in comparison with classical RDHEI.

The comparative analysis in Tables 3 and 4 reveals distinct embedding capacity characteristics between the two methods. For the parameter configuration $\beta = 5$, $n_b = 4$, $T = 4$ in PQ-RDHEI versus $\beta = 5$, $n = 8$, $T = 4$ in Saeidi & Mirzakuchaki, the pure capacity ratios range from a factor of 17.6 for Airplane (595,728 bits versus 33,786 bits) to 4.8 for Baboon (160,190 bits versus 33,896 bits). This variation correlates with image texture complexity, as images with smoother regions (Airplane, Peppers, Jet) admit substantially higher capacities under the $n_b = 4$ block structure, whereas highly textured images (Baboon) exhibit reduced embeddable block counts. The embedding rate metric quantifies this precisely: PQ-RDHEI achieves 2.2725 bits per pixel for Airplane compared to 0.1289 bpp in the reference method, while Baboon yields 0.6111 bpp versus 0.1293 bpp. The entropy analysis in Table 5 demonstrates that PQ-RDHEI encrypted shares attain entropy values approximating the theoretical maximum of 8 bits (ranging from 7.9902 to 7.9992), whereas the reference method produces shares with entropy near 7.48 bits. This 0.5-bit entropy increase per pixel indicates enhanced statistical randomness in the encrypted domain, directly attributable to the ML-KEM-based KEM-DEM encryption structure versus the stream cipher-only approach.

Table 5 Entropy analysis ($\beta = 5$, $T = 4$), comparison with Saeidi & Mirzakuchaki [24]

	Saeidi & Mirzakuchaki [24]		Our PQ-RDHEI		
Image	Share (avg)	Re-const. Image	Share (avg)	Re-const. Image	Original Image
Airplane	7.48	7.4887	7.9972	5.4474	5.4474
Baboon	7.48	7.3002	7.9991	7.3743	7.3743
Barbara	7.48	7.6029	7.9992	7.5674	7.5674
Boat	7.48	7.1996	7.9986	7.2095	7.2095
Cameraman	7.48	7.1063	7.9902	7.1045	7.1045
Jet	7.48	7.6184	7.9976	6.7240	6.7240
Lake	7.72	7.4887	7.9985	7.4758	7.4758
Peppers	7.48	7.6184	7.9983	7.5865	7.5865

Table 6 Embedding rates (bpp) for block size 2×2, comparison with Saeidi & Mirzakuchaki [24]

	Saeidi & Mirzakuchaki [24]				Our PQ-RDHEI			
Image	β=3	β=4	β=5	β=6	β=3	β=4	β=5	β=6
	T=2	T=2	T=2	T=2	T=2	T=2	T=2	T=2
Airplane	1.6845	1.9937	0.9590	1.0300	0.9888	1.9359	2.7906	3.4341
Baboon	1.1642	1.2345	1.4713	0.8464	0.7781	1.0736	0.9295	0.5629
Barbara	1.3153	1.3759	1.4573	0.8928	0.8888	1.4806	1.7143	1.4427
Boat	1.4497	1.3810	1.5806	0.8937	0.9573	1.7217	1.9830	1.6838
Cameraman	1.5780	1.7120	1.7682	0.9869	0.9276	1.6462	2.0755	2.3591
Jet	1.5864	1.5708	1.5082	0.9468	0.9629	1.7919	2.3645	2.5676
Lake	1.3265	1.4186	1.5917	0.9030	0.9355	1.6145	1.8513	1.6702
Peppers	1.1954	1.3616	1.5384	0.8841	0.9801	1.8626	2.4642	2.4403

We stress that entropy is used here strictly as a statistical indicator and carries no cryptographic meaning on its own. First-order Shannon entropy close to the maximum of 8 bits per pixel establishes only that the marginal distribution of the encrypted pixel values is close to uniform; it is a necessary but far from sufficient condition, and it is satisfied by constructions that are trivially insecure, for example a fixed one-time pad reused across images, or a permutation-only cipher, which preserves the histogram exactly. High entropy consequently does not establish semantic security, indistinguishability under chosen-plaintext or chosen-ciphertext attack, or resistance to any specific cryptanalysis, and no such claim is made in this paper. The cryptographic guarantees of PQ-RDHEI come exclusively from the reduction-based theorems of Section 6, semantic security in Theorem 3, payload privacy in Theorem 4, and robustness in Theorem 7, which hold against QPT adversaries in the QROM and would remain valid even if the measured entropy were substantially lower. The entropy figures are reported only because they are the conventional descriptive statistic in the RDHEI literature and permit comparison with prior work on that literature's own terms.

Tables 6 and 7 quantify the embedding rate dependence on block geometry and MSB parameter selection. For 2×2 blocks with $\beta = 5$, $T = 2$, PQ-RDHEI embedding rates span 0.9295 bpp (Baboon) to 2.7906 bpp (Airplane), contrasting with the reference range of 0.9590 bpp to 1.7682 bpp under comparable conditions. The 4×4 block configuration reduces maximum rates to 0.1298 bpp (Baboon, $\beta = 5, T = 2$) through 1.9287 bpp (Peppers, $\beta = 5, T = 3$) for PQ-RDHEI, compared to 0.3589 bpp through 0.4160 bpp in the reference

Table 7 Embedding rates (bpp) for block size 4×4, comparison of PQ-RDHEI with Saeidi & Mirzakuchaki [24]

	Saeidi & Mirzakuchaki [24]								Our PQ-RDHEI							
Image	β=3	β=3	β=4	β=4	β=5	β=5	β=6	β=6	β=3	β=3	β=4	β=4	β=5	β=5	β=6	β=6
	T=2	T=3	T=2	T=3	T=2	T=3	T=2	T=3	T=2	T=3	T=2	T=3	T=2	T=3	T=2	T=3
Airplane	0.3444	0.4327	0.3108	0.4845	0.3609	0.4138	0.2798	0.5908	0.9490	0.9677	1.8143	1.8702	1.9708	2.1614	1.5737	2.1313
Baboon	0.2203	0.2344	0.2500	0.2509	0.3589	0.3589	0.2803	0.5905	0.4030	0.5347	0.3794	0.6325	0.1298	0.3871	0.0122	0.0865
Barbara	0.2458	0.2870	0.2560	0.3108	0.3598	0.3642	0.2806	0.5905	0.6344	0.7357	0.9299	1.1325	0.7744	1.2262	0.2350	0.7354
Boat	0.2472	0.2874	0.2608	0.2846	0.3609	0.3716	0.2801	0.5915	0.7711	0.8465	1.0441	1.3791	0.7727	1.2732	0.3804	0.7779
Cameraman	0.2000	0.3503	0.3106	0.2922	0.3652	0.4160	0.2804	0.5921	0.7524	0.8202	1.2090	1.3778	1.4226	1.5795	1.2804	1.5129
Jet	0.2881	0.3399	0.2875	0.3398	0.3675	0.4029	0.2798	0.5948	0.8212	0.8742	1.3621	1.5489	1.4547	1.7579	1.0981	1.5221
Lake	0.2531	0.2995	0.2650	0.2922	0.3602	0.3779	0.2814	0.5908	0.6999	0.7944	1.0110	1.2666	0.8871	1.2962	0.5732	0.9647
Peppers	0.2400	0.2898	0.2517	0.2718	0.3587	0.3595	0.2803	0.5900	0.8726	0.9177	1.4456	1.6516	1.3640	1.9287	0.6633	1.3555



(a) Airplane (b) Baboon

(c) Barbara (d) Boat

(e) Cameraman (f) Jet

(g) Lake (h) Peppers

Fig. 3 Histogram analysis of test images: original cover images (left) and their corresponding histogram comparisons (right) showing pixel distribution before and after extraction. The histogram comparisons demonstrate that the proposed PQ-RDHEI scheme achieves perfect reconstruction with pixel distributions of extracted images identical to the original images

method. This block size sensitivity reflects the fundamental trade-off between spatial correlation preservation and embeddable region identification. The histogram analysis in Fig. 3 provides distributional validation: pixel value distributions in recovered images exhibit exact correspondence with original distributions, confirming bit-exact reconstruction. The concentration of histogram mass at intensity values 123, 127, 120, and 125 in the Airplane example persists unchanged through the encryption-embedding-extraction-decryption pipeline, verifying that the reversibility property holds at the pixel-level distribution granularity rather than merely at aggregate statistical measures.

5.1 Runtime measurement and reproducibility

Regarding computational overhead, PQ-RDHEI incurs a small additional runtime with respect to a classical stream-cipher RDHEI pipeline, primarily due to ML-KEM operations and Π_{LA} verification. To make this statement reproducible we report the measurement protocol in full. All timings were obtained on an Apple M2 system-on-chip (8 cores, 4 performance and 4 efficiency) with 16 GiB of unified memory, running macOS 26.5, with the machine otherwise idle and no other user processes active. The implementation is written in Python 3.12.3 with NumPy 2.3.1; all image-domain operations, block partitioning, mask application, MSB analysis, Huffman encoding and decoding, and the bit-level interleaving of WriteEncoded and ParseMSB, are vectorised as array operations, so that the only interpreter-level loops are $O(\ell)$ in the number of blocks. ML-KEM-768 is instantiated with a portable reference implementation of FIPS 203, SHAKE256 and SHA3-256 are taken from the platform `hashlib`, and Π_{LA} uses a (1, 2) threshold over the Mersenne prime field $\mathbb{Z}_{2^{127}-1}$ with SHA3-256 commitments. Each reported figure is the mean $\bar{T}$ over 30 independent runs preceded by 2 discarded warm-up runs; each run performs a fresh KeyGen, Enc, Embed, Extract and Dec on the 512×512 Jet image with a payload filling half of the net capacity, and every run was verified to recover the payload, the ciphertext and the image bit-exactly.

Table 8 reports $\bar{T} \pm \sigma$ together with the half-width of the 95% confidence interval $1.96\,\sigma/\sqrt{n}$ and the observed range. The complete pipeline takes 91.66 ± 6.85 ms, so the confidence interval on the mean is ± 2.45 ms, or 2.7% of the mean; the distribution is right-skewed, with a median close to the minimum and a small number of slower runs attributable to scheduler interference, which is why we report the range as well as the standard deviation. Extraction dominates at 50.24 ms because it must re-derive the Huffman decoding of every marked block and re-verify both the commitment and the tag, whereas encryption and decryption are essentially the cost of one ML-KEM operation plus one pass of the stream cipher. Table 8 also isolates the components that a classical RDHEI scheme would not pay for: ML-KEM.Encaps and ML-KEM.Decaps together with Π_{LA}.Share and Π_{LA}.VerifyShare account for 7.49 ± 0.22 ms, that is 8.2% of the pipeline, which quantifies the price of post-quantum security and of payload integrity in this construction. The Π_{LA} contribution is

Table 8 Runtime of PQ-RDHEI on the 512×512 Jet image over $n = 30$ independent runs (2 warm-up runs discarded), on an Apple M2 with 16 GiB, macOS 26.5, Python 3.12.3 and NumPy 2.3.1. CI denotes the half-width $1.96\,\sigma/\sqrt{n}$ of the 95% confidence interval on the mean

Phase	$\bar{T}$ (ms)	σ (ms)	95% CI (ms)	Min (ms)	Max (ms)
Enc	9.26	0.22	±0.08	9.0	9.8
Embed	21.66	0.97	±0.35	20.7	25.0
Extract	50.24	5.79	±2.07	47.1	78.3
Dec	10.50	0.50	±0.18	10.1	12.5
Total pipeline	**91.66**	**6.85**	±52.45	**87.2**	**124.2**
Post-quantum specific components (included in the phases above)					
ML-KEM.KeyGen	2.367	0.082	±0.029	—	—
ML-KEM.Encaps	3.172	0.169	±0.060	—	—
ML-KEM.Decaps	4.283	0.140	±0.050	—	—
Π_{LA}.Share	0.019	0.005	±0.002	—	—
Π_{LA}.VerifyShare + Reconstruct	0.011	0.001	±0.001	—	—
Post-quantum subtotal	**7.49**	**0.22**	±0.08	**8.2% of pipeline**	

negligible at 0.03 ms, since a (1, 2) threshold requires two polynomial evaluations and three hash calls; effectively all of the overhead is the key encapsulation, and it is a fixed per-image cost that does not grow with the image, so its relative weight falls as resolution increases, at 4096×4096 it is below 0.11% of the pipeline.

5.2 Evaluation on medical images

The evaluation above, like most of the RDHEI literature, uses the USC-SIPI natural images. Since long-term confidentiality of medical imagery is one of the motivating applications of Section 1, we repeated the evaluation on a medical test set covering four acquisition modalities: computed tomography, magnetic resonance, ultrasound and projection radiography. The eight images were drawn from the MedPix collection served by the Open-i service of the U.S. National Library of Medicine [38], which is openly accessible and requires no registration, and each was converted to 8-bit grayscale, cropped to its largest centred square and scaled to 512×512 so that the figures are directly comparable with Table 3. Parameters are unchanged at $\beta = 5$, $n_b = 4$, $T = 4$.

Table 9 reports the results. The capacity advantage is not merely preserved on medical data, it improves: the net embedding rate ranges from 1.1648 bpp for the sagittal spine MRI to 1.9921 bpp for the hand radiograph, with a mean of 1.5738 bpp, against a mean of 1.1464 bpp for the eight standard natural images under an identical accounting, and against 0.043–0.177 bpp for the reference method of Table 3. Even the least favourable medical image therefore exceeds the reference method's best natural image by a factor of 6.6. The reason is structural rather than incidental: radiographic and tomographic images contain large collimated or air-filled regions of near-constant intensity, so the proportion of blocks with $\alpha_k \le T$ distinct MSB patterns is high, between 63.8% and 94.4% of blocks are embeddable, compared with 24.5% for Baboon, and it is precisely this statistic that the capacity function of Definition 15 rewards. The correlation is visible in the table: the two images with the lowest plaintext entropy, the chest CT at 3.5714 bits and the thyroid ultrasound at 5.4150 bits, are not the highest-capacity images, because what matters is intra-block MSB homogeneity rather than global histogram concentration; the hand and chest radiographs

Table 9 PQ-RDHEI on 512×512 medical images from four modalities ($\beta = 5$, $n_b = 4$, $T = 4$). Entire capacity is $\sum_{k \in \mathcal{E}} \mathsf{cap}_k$; pure capacity is net of the location map, the per-block MSB tables, the VSS proof and the length fields; the rate is pure capacity per pixel. H_{plain} and H_{share} are the entropies of the cover image and of the encrypted share

Image	Modality	Entire Cap. (bits)	Pure Cap. (bits)	Rate (bpp)	H_{plain}	H_{share}	PSNR (dB)	SSIM
CT-Abdomen	CT	465830	324934	1.2395	5.5388	7.9948	$+\infty$	1.000000
CT-Chest	CT	614265	480472	1.8329	3.5714	7.9930	$+\infty$	1.000000
MRI-Brain	MRI	550336	377171	1.4388	7.0687	7.9968	$+\infty$	1.000000
MRI-Spine	MRI	542598	305347	1.1648	6.8677	7.9988	$+\infty$	1.000000
US-Abdomen	Ultrasound	513651	315634	1.2040	6.7063	7.9972	$+\infty$	1.000000
US-Thyroid	Ultrasound	651624	471280	1.7978	5.4150	7.9956	$+\infty$	1.000000
XR-Chest	Radiograph	711455	503431	1.9204	7.0767	7.9970	$+\infty$	1.000000
XR-Hand	Radiograph	711970	522218	1.9921	6.6159	7.9958	$+\infty$	1.000000
Mean (medical)		595216	412561	**1.5738**	6.1076	7.9961	$+\infty$	1.000000
Mean (natural, same accounting)		410370	268266	1.1464	—	7.9972	$+\infty$	1.0

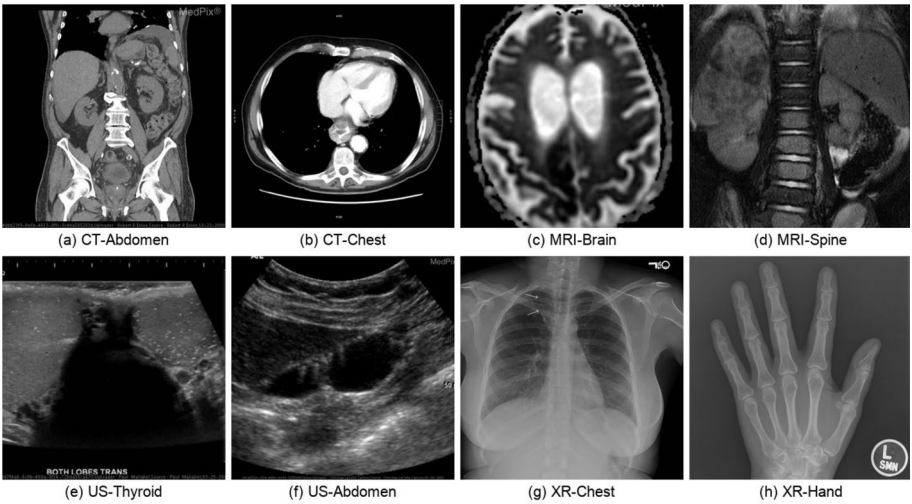


Fig. 4 The medical test set used in Table 9: eight 512×512 grayscale images spanning computed tomography, magnetic resonance, ultrasound and projection radiography, taken from the openly accessible MedPix collection of the U.S. National Library of Medicine. All eight are recovered bit-exactly by PQ-RDHEI

combine moderate global entropy with very smooth local structure and yield the highest rates. Encrypted-share entropy remains between 7.9930 and 7.9988 bits per pixel, indistinguishable from the natural-image case as expected, since the encryption layer is independent of image content, and every image was recovered bit-exactly, giving $\mathrm{PSNR} = +\infty$ and $\mathrm{SSIM} = 1$ throughout. Figure 4 shows the test set.

5.3 Scalability

Table 10 reports the full pipeline at five resolutions from 256×256 to 4096×4096, a 256-fold range in pixel count. To keep image statistics comparable across scales, resampling a single image would change block homogeneity and therefore capacity, confounding the measurement, each canvas is a mosaic tiled from the eight standard 512×512 test images at their native resolution, so that local statistics are those of natural images at every scale; the 256×256 entry is a crop of the Jet image. Timings are means over five runs in a fresh process per resolution, and peak memory is the traced peak of the allocator over one complete pipeline, which is independent of the operating system's resident-set accounting.

The measurements confirm linear scaling on both axes. Throughput is essentially flat at 0.345–0.453 µs per pixel across the whole range, with the two extremes explained by fixed costs at the small end, where the 7.49 ms of ML-KEM and Π_{LA} work is 25% of a 29.7 ms pipeline, and by cache pressure at the large end, where the working set exceeds the last-level cache; between 512×512 and 2048×2048 the variation is under 8%. Peak memory is likewise linear at a constant 96.0 MiB per megapixel from 512×512 upwards, which is the expected cost of holding the block array together with the per-block symbol histogram and the rank, code-length and capacity arrays; the 256×256 figure of 98.6 MiB per megapixel is marginally higher only because of the fixed-size buffers. Capacity grows linearly as well,

Table 10 Scalability of PQ-RDHEI ($\beta = 5$, $n_b = 4$, $T = 4$). Each canvas is a mosaic of the eight standard 512×512 test images at native resolution; the 256×256 entry is a crop of Jet. Timings are $\bar{T} \pm \sigma$ over five runs per resolution in a fresh process, on the platform of Table 8. Peak memory is the traced allocator peak over one complete pipeline

Size	Pixels	Enc (ms)	Embed (ms)	Extract (ms)	Dec (ms)	Total (ms)	Through-put (µs/px)	Peak mem. (MiB)	Entire cap. (bits)
256×256	65 536	4.8 ± 0.1	5.6 ± 0.1	13.4 ± 1.5	5.8 ± 0.1	29.7 ± 1.5	0.453	6.5	159 925
512×512	262 144	9.3 ± 0.1	21.7 ± 0.3	52.4 ± 3.4	10.4 ± 0.0	93.8 ± 3.3	0.358	25.2	547 026
1024×1024	1 048 576	28.2 ± 0.2	92.2 ± 2.4	211.9 ± 11.5	29.3 ± 0.3	361.6 ± 11.7	0.345	100.6	1 845 449
2048×2048	4 194 304	106.4 ± 1.9	403.0 ± 8.2	948.1 ± 34.7	109.5 ± 2.6	1566.9 ± 39.3	0.374	402.6	7 493 900
4096×4096	16 777 216	463.2 ± 25.0	1843.3 ± 249.0	4588.1 ± 553.4	488.0 ± 32.4	7382.6 ± 797.2	0.440	1610.2	29 975 600

from 159 925 bits to 29 975 600 bits, with the embedding rate settling at 1.7867 bpp once the mosaic contains the full set of eight tiles. Extraction remains the dominant phase at all scales, at 45–62% of the pipeline. The practical consequence is that a 4096×4096 image, the largest supported by the parameter choice $\ell \leq 2^{20}$ of Section 4, completes in 7.4 s and 1.6 GiB in an interpreted implementation, so the scheme is not limited by scale in the resolutions typical of medical and archival imaging.

6 Security and privacy analysis of PQ-RDHED

Here, we formally address that our technique is correct (Theorems 1 and 2), compact (Theorem 6), semantically secure (Theorem 3), robust (Theorem 7), and provides payload privacy (Theorem 4) as well as separability (Theorem 5).

Remark 6 (Key relationship and security). For correct operation of PQ-RDHED, the hiding key $hk = K_h$ and extraction key $xk = K_x$ satisfy the constraint $K_h = K_x$. This enables the data hider to embed payloads that can be correctly extracted by the receiver. In the key distribution model: (ek, dk) are held by the content owner who encrypts images; (hk, xk) with $hk = xk$ are distributed such that the data hider receives hk for embedding and the receiver obtains xk for extraction. Under this constraint, payload privacy reduces to semantic security of the underlying encryption, as any party with hk can derive the masking keys. For applications requiring stronger payload privacy with $hk \neq xk$, the construction can be extended by including an additional public-key encryption layer for the payload under a key held only by the receiver.

Since this constraint is the one structural concession of the construction, we state its consequences precisely. Table 11 lists which entity holds which key and what it can therefore learn. The content owner holds (ek, dk) and learns the image but, lacking hk, learns nothing about the payload beyond its length; the data hider and the receiver share the single value $hk = xk$, from which both derive $K_{\mathsf{pos}} = \mathsf{H}(hk||\mathsf{0x02})$ and $K_{\mathsf{mask}} = \mathsf{H}(hk||\mathsf{0x03})$, and neither learns anything about the image, as Embed and Extract never invoke dk. The separation that $hk = xk$ weakens is therefore *between the hider and the receiver*, not between either of them and the image: an entity able to extract is also able to embed and to forge tag , so the authentication provided by tag is a message authentication code shared by the two parties rather than a signature that distinguishes them. If hk is compromised, the adversary obtains payload confidentiality (it unmasks $\tilde{d}$) and payload integrity (it re-embeds arbitrary payloads with a valid tag), whereas image confidentiality is untouched, being reducible to

Table 11 Key possession and knowledge under the standard key model $hk = xk$. "Image" and "Payload" state what the entity can recover; $\hat{c}$ denotes a marked ciphertext

Entity	Keys held	Image	Payload	Can forge tag
Content owner	ek, dk	✓	✗	✗
Data hider	hk	✗	✓	✓
Receiver	$xk\,(= hk)$	✗	✓	✓
Receiver (full)	dk, xk	✓	✓	✓
Storage provider / eavesdropper	—	✗	✗	✗

the IND-CCA security of ML-KEM under dk; a compromised hk can be rotated without re-encrypting the image, since Embed operates purely in the encrypted domain. Payload confidentiality therefore remains meaningful under the standard key model against every party that does *not* hold hk, in particular against the storage provider, the content owner and any network observer, which is the adversary of Remark 3, and Theorem 4 bounds exactly this. We keep $hk = xk$ as the default because it preserves strong compactness in the sense of Definition 12 up to the λ-bit tag: the extended model needs a per-image public-key ciphertext of the payload key, which enlarges $\hat{c}$ by $|c_{\mathsf{kem}}| = 1088$ bytes and requires the receiver to run a second decapsulation, so we present it as an optional layer rather than as the baseline.

Theorem 1 *Our proposed* PQ-RDHED *is correct (Definition 7); specifically, for all $\lambda \in \mathbb{N}$, $m \in \mathcal{M}$, $d \in \mathcal{D}$ with $|d| \le \kappa(|m|)$:*

$$\Pr\left[\begin{array}{l}(ek, dk, hk, xk) \leftarrow KeyGen(1^\lambda);\ c \leftarrow Enc(ek, m);\\ \hat{c} \leftarrow Embed(hk, c, d);\ (c', d') \leftarrow Extract(xk, \hat{c})\end{array} : c' = c \wedge d' = d\right] = 1.$$

Proof Let $(ek, dk, hk, xk) \leftarrow \mathsf{KeyGen}(1^\lambda)$ with $hk = K_h$ and $xk = K_x$ satisfying $K_h = K_x$ per Remark 6. Let $c = (c_{\mathsf{kem}}, \{E'_k\}_{k=1}^{\ell}) \leftarrow \mathsf{Enc}(ek, m)$ and $\hat{c} = (c_{\mathsf{kem}}, \{\hat{E}_k\}_{k=1}^{\ell}, \mathsf{tag}) \leftarrow \mathsf{Embed}(hk, c, d)$. Since $K_h = K_x$, the derived keys satisfy:

$$K_{\mathsf{pos}}^{(\mathsf{emb})} = \mathsf{H}(K_h \| 0\mathrm{x}02) = \mathsf{H}(K_x \| 0\mathrm{x}02) = K_{\mathsf{pos}}^{(\mathsf{ext})},$$
$$K_{\mathsf{mask}}^{(\mathsf{emb})} = \mathsf{H}(K_h \| 0\mathrm{x}03) = \mathsf{H}(K_x \| 0\mathrm{x}03) = K_{\mathsf{mask}}^{(\mathsf{ext})}.$$

For each embeddable block $k \in \mathcal{E}$ with $\alpha_k \le T$ distinct MSB types, define the encoding map $\phi_k : \{0, \ldots, 2^\beta - 1\}^{n_b^2} \times \{0,1\}^{\mathsf{cap}_k} \to \{0,1\}^{\beta \cdot n_b^2}$ by:

$$\phi_k\left(\{\mathsf{msb}_{k,u}\}_{u=1}^{n_b^2}, \mathsf{emb}_k\right) = \bigcup_{u=1}^{n_b^2} \left((0)^{p_{k,u}} \| 1 \| \overline{\mathsf{code}_k(\mathsf{msb}_{k,u})}\right),$$

where $\sum_{u=1}^{n_b^2} p_{k,u} = \mathsf{cap}_k$ and $\overline{\cdot}$ denotes bit reversal. The delimiter bit '1' ensures prefix-freeness, hence ϕ_k is injective. Since domain and codomain have equal cardinality $2^{\beta \cdot n_b^2}$, ϕ_k is bijective with inverse ϕ_k^{-1} computable in $O(n_b^2)$ time. The embedding constructs $\mathsf{stream} = \mathsf{len}(d) \| \tilde{d} \| \mathsf{aux} \| \pi_{\mathsf{share}}$ where $\tilde{d} = d \oplus \mathsf{PRF}(K_{\mathsf{mask}}, 0\mathrm{x}00)[0 : |d|]$. Extraction recovers $\mathsf{stream}' = \mathsf{len}(d') \| \tilde{d}' \| \mathsf{aux}' \| \pi'_{\mathsf{share}}$ by applying ϕ_k^{-1} to each block $k \in \mathcal{E}'$ in lexicographic order. By bijectivity of ϕ_k and determinism of block ordering $\mathsf{stream}' = \mathsf{stream} \implies (\mathsf{len}(d'), \tilde{d}', \mathsf{aux}', \pi'_{\mathsf{share}}) = (\mathsf{len}(d), \tilde{d}, \mathsf{aux}, \pi_{\mathsf{share}})$. From the aforementioned steps of this proof, we have:

$$d' = \tilde{d}' \oplus \mathsf{PRF}(K_{\mathsf{mask}}^{(\mathsf{ext})}, 0\mathrm{x}00)[0 : |\tilde{d}'|] = \tilde{d} \oplus \mathsf{PRF}(K_{\mathsf{mask}}^{(\mathsf{emb})}, 0\mathrm{x}00)[0 : |\tilde{d}|] = d.$$

The auxiliary data $\mathsf{aux} = L \| |\mathcal{E}| \| (\alpha_k, \{\mathsf{msb}_k^{(j)}\}_{j=1}^{\alpha_k})_{k \in \mathcal{E}}$ encodes the location map and original MSB metadata. For each k with $L[k] = 1$, reconstruction applies: $E'_k[u] = \left(\mathsf{msb}_{k,u} \cdot 2^{8-\beta}\right) + \left(\hat{E}_k[u] \mod 2^{8-\beta}\right)$, $\forall u \in [n_b^2]$. Since $\mathsf{msb}_{k,u}$ is recovered exactly via ϕ_k^{-1} and LSB positions are unmodified by embedding, $E'_k = E_k$ for all $k \in [\ell]$. Thus $c' = (c_{\mathsf{kem}}, \{E'_k\}_{k=1}^{\ell}) = c$. □

Theorem 2 PQ-RDHED *satisfies decryption correctness (Definition 8). For all* $\lambda \in \mathbb{N}$, $m \in \mathcal{M}$:

$$\Pr\left[(ek, dk, hk, xk) \leftarrow KeyGen(1^\lambda);\ c \leftarrow Enc(ek, m) : Dec(dk, c) = m\right] \geq 1 - \delta_{kem},$$

where $\delta_{kem} = negl(\lambda)$ *denotes the decapsulation failure probability of* ML-KEM.

Proof of Sketch Let $(pk, sk) \leftarrow \mathsf{ML\text{-}KEM.KeyGen}(1^\lambda)$ with $ek = pk$, $dk = sk$. Let $c = (c_{\mathsf{kem}}, \{E'_k\}_{k=1}^{\ell}) \leftarrow \mathsf{Enc}(ek, m)$where$(c_{\mathsf{kem}}, K) \leftarrow \mathsf{ML\text{-}KEM.Encaps}(pk)$.Bythe$(1-\delta_{\mathsf{kem}})$-correctness of ML-KEM: $\Pr[K' = K \mid K' \leftarrow \mathsf{ML\text{-}KEM.Decaps}(sk, c_{\mathsf{kem}})] \geq 1 - \delta_{\mathsf{kem}}$. Define event $\mathcal{E}_{\mathsf{kem}} := \{K' = K\}$. We condition on $\mathcal{E}_{\mathsf{kem}}$ for the remainder. Conditioned on $\mathcal{E}_{\mathsf{kem}}$, Dec computes $K'_e = \mathsf{H}(K'\|\mathsf{0x00}) = \mathsf{H}(K\|\mathsf{0x00}) = K_e$, $K'_\pi = \mathsf{H}(K'\|\mathsf{0x01}) = \mathsf{H}(K\|\mathsf{0x01}) = K_\pi$. Let $\sigma = \mathsf{FisherYates}(\ell, K_\pi) : [\ell] \to [\ell]$. Since FisherYates is deterministic given (ℓ, K_π) and produces a bijection, σ^{-1} exists and is computable. By construction: $E'_{\sigma(k)} = E_k$ during encryption, hence $E_k = E'_{\sigma^{-1}(\sigma(k))} = E'_k \circ \sigma^{-1}$. For each $k \in [\ell]$, let $E_k = \{e_{k,u}\}_{u=1}^{n_b^2}$ with $e_{k,u} = p_{k,u} \oplus \mu_k$ where $\mu_k = \mathsf{PRF}(K_e, k)[0:8]$. Decryption computes $\mu'_k = \mathsf{PRF}(K'_e, k)[0:8] = \mathsf{PRF}(K_e, k)[0:8] = \mu_k$ and $p'_{k,u} = e_{k,u} \oplus \mu'_k = (p_{k,u} \oplus \mu_k) \oplus \mu_k = p_{k,u}$, $\forall u \in [n_b^2]$. The partition $\{B_k\}_{k=1}^{\ell}$ of m into $n_b \times n_b$ blocks is deterministic given (W, H, n_b). Since $B'_k = \{p'_{k,u}\}_{u=1}^{n_b^2} = \{p_{k,u}\}_{u=1}^{n_b^2} = B_k$ for all k, reconstruction yields $m' = m$. Eventually, we will have the probability bound:

$$\Pr[\mathsf{Dec}(dk, c) = m] \geq \Pr[\mathcal{E}_{\mathsf{kem}}] \cdot \Pr[\mathsf{Dec}(dk, c) = m \mid \mathcal{E}_{\mathsf{kem}}] = (1 - \delta_{\mathsf{kem}}) \cdot 1 = 1 - \delta_{\mathsf{kem}}.$$

□

Theorem 3 PQ-RDHED *satisfies semantic security (Definition 9) in the QROM. For any QPT adversary* $\mathcal{A}$ *making at most* q_H *quantum queries to* H *and* q_D *classical decapsulation queries:* $Adv^{sem}_{\mathsf{PQ\text{-}RDHED},\mathcal{A}}(\lambda) \leq Adv^{\mathsf{IND\text{-}CCA}}_{\mathsf{ML\text{-}KEM},\mathcal{B}_1}(\lambda) + \sum_{k=1}^{\ell} Adv^{PRF}_{PRF,\mathcal{B}_{2,k}}(\lambda)$, *where* $\mathcal{B}_1$ *is a QPT reduction to IND-CCA security of* ML-KEM *and* $\{\mathcal{B}_{2,k}\}_{k=1}^{\ell}$ *are QPT reductions to PRF security.*

Proof We proceed via a sequence of games $G_0, G_1, \ldots, G_{\ell+1}$.

Game G_0 Define $p_0 := \Pr[G_0 \text{ outputs } 1]$. The real experiment $\mathsf{Expt}^{\mathsf{sem}}_{\mathcal{A},b}(\lambda)$:

1. $(pk, sk) \leftarrow \mathsf{ML\text{-}KEM.KeyGen}(1^\lambda)$; $K_h \leftarrow \{0,1\}^\lambda$; $K_x \leftarrow \{0,1\}^\lambda$.
2. $\mathcal{A}(pk, K_h, K_x)$ outputs (m_0, m_1) with $|m_0| = |m_1|$.
3. $(c_{\mathsf{kem}}, K) \leftarrow \mathsf{ML\text{-}KEM.Encaps}(pk)$; $K_e \leftarrow \mathsf{H}(K\|\mathsf{0x00})$; $K_\pi \leftarrow \mathsf{H}(K\|\mathsf{0x01})$.
4. For $k \in [\ell]$: $\mu_k \leftarrow \mathsf{PRF}(K_e, k)[0:8]$; $E_k \leftarrow \{m_b[k,u] \oplus \mu_k\}_{u=1}^{n_b^2}$.
5. $\sigma \leftarrow \mathsf{FisherYates}(\ell, K_\pi)$; $c^* \leftarrow (c_{\mathsf{kem}}, \{E_{\sigma^{-1}(k)}\}_{k=1}^{\ell})$.
6. $b' \leftarrow \mathcal{A}(c^*)$; output b'.**Game** G_1 Identical to G_0 except: replace $(c_{\mathsf{kem}}, K) \leftarrow \mathsf{ML\text{-}KEM.Encaps}(pk)$ with $(c_{\mathsf{kem}}, K) \leftarrow \mathsf{ML\text{-}KEM.Encaps}(pk)$ but substitute $K \leftarrow \{0,1\}^{256}$ uniformly at random (i.e., K is independent of c_{kem}). Define $p_1 := \Pr[G_1 \text{ outputs } 1]$.

Claim 1 $|p_0 - p_1| \leq \mathsf{Adv}^{\text{IND-CCA}}_{\text{ML-KEM},\mathcal{B}_1}(\lambda)$.

Proof of Claim 1 Construct $\mathcal{B}_1$ as follows: $\mathcal{B}_1$ receives $(pk, c^*_{\mathsf{kem}}, K^*)$ from the IND-CCA challenger where K^* is either the real encapsulated key or uniform random. $\mathcal{B}_1$ sets $c_{\mathsf{kem}} := c^*_{\mathsf{kem}}$, $K := K^*$, simulates the remainder of the game, and outputs $\mathcal{A}$'s guess. If K^* is real, $\mathcal{B}_1$ simulates G_0; if K^* is random, $\mathcal{B}_1$ simulates G_1. Thus:

$$|p_0 - p_1| = \left|\Pr[\mathcal{B}_1^{\mathcal{O}_{\text{dec}}}(\text{real}) = 1] - \Pr[\mathcal{B}_1^{\mathcal{O}_{\text{dec}}}(\text{rand}) = 1]\right| = \mathsf{Adv}^{\text{IND-CCA}}_{\text{ML-KEM},\mathcal{B}_1}(\lambda).$$

Games G_{1+k} **for** $k \in [\ell]$ ***Identical to*** G_k ***except: replace*** $\mu_k = \mathsf{PRF}(K_e, k)[0:8]$ ***with*** $\mu_k \leftarrow \{0, \ldots, 255\}$ ***sampled uniformly at random. Define*** $p_{1+k} := \Pr[G_{1+k} \text{ outputs } 1]$**.**

Claim 2 For each $k \in [\ell]$: $|p_k - p_{1+k}| \leq \mathsf{Adv}^{\text{PRF}}_{\text{PRF},\mathcal{B}_{2,k}}(\lambda)$.

Proof of Claim 2 Construct $\mathcal{B}_{2,k}$ with oracle access to $\mathcal{O}$ (either $\mathsf{PRF}(K_e, \cdot)$ or a random function). For indices $j < k$, $\mathcal{B}_{2,k}$ samples $\mu_j \leftarrow \{0, \ldots, 255\}$. For index k, $\mathcal{B}_{2,k}$ queries $\mu_k \leftarrow \mathcal{O}(k)[0:8]$. For indices $j > k$, $\mathcal{B}_{2,k}$ queries $\mu_j \leftarrow \mathcal{O}(j)[0:8]$. If $\mathcal{O} = \mathsf{PRF}(K_e, \cdot)$, this simulates G_k; if $\mathcal{O}$ is random at position k, this simulates G_{1+k}. Thus $|p_k - p_{1+k}| \leq \mathsf{Adv}^{\text{PRF}}_{\text{PRF},\mathcal{B}_{2,k}}(\lambda)$.

To analyze $G_{\ell+1}$, all masks $\{\mu_k\}_{k=1}^{\ell}$ are uniformly random and independent. For each block k and pixel position u: $E_k[u] = m_b[k,u] \oplus \mu_k$, where $\mu_k \leftarrow \{0, \ldots, 255\}$ is independent of m_b. By the XOR-lemma, $E_k[u]$ is uniformly distributed over $\{0, \ldots, 255\}$ independent of $m_b[k,u]$. Since the permutation σ is determined by $K_\pi = \mathsf{H}(K \| 0\text{x}01)$ with K uniform random (independent of b), the permuted blocks $\{E_{\sigma^{-1}(k)}\}_{k=1}^{\ell}$ are identically distributed for $b = 0$ and $b = 1$. Hence:

$$p_{\ell+1} = \Pr[G_{\ell+1} \text{ outputs } 1 \mid b = 0] = \Pr[G_{\ell+1} \text{ outputs } 1 \mid b = 1] \implies \Pr[b' = b] = \frac{1}{2}.$$

To calculate the advantage, we will have:

$$\begin{aligned}
\mathsf{Adv}^{\mathsf{sem}}_{\text{PQ-RDHED},\mathcal{A}}(\lambda) &= |\Pr[b' = 1 \mid b = 0] - \Pr[b' = 1 \mid b = 1]| = 2\left|p_0 - \frac{1}{2}\right| \\
&\leq 2\left(|p_0 - p_1| + \sum_{k=1}^{\ell} |p_k - p_{1+k}| + \left|p_{\ell+1} - \frac{1}{2}\right|\right) \\
&\leq 2\left(\mathsf{Adv}^{\text{IND-CCA}}_{\text{ML-KEM},\mathcal{B}_1}(\lambda) + \sum_{k=1}^{\ell} \mathsf{Adv}^{\text{PRF}}_{\text{PRF},\mathcal{B}_{2,k}}(\lambda)\right) \\
&= 2 \cdot \mathsf{Adv}^{\text{IND-CCA}}_{\text{ML-KEM},\mathcal{B}_1}(\lambda) + 2\sum_{k=1}^{\ell} \mathsf{Adv}^{\text{PRF}}_{\text{PRF},\mathcal{B}_{2,k}}(\lambda).
\end{aligned}$$

□

Theorem 4 PQ-RDHED *satisfies payload privacy (Definition 10) in the QROM under the extended key model where* $hk \neq xk$. *For any QPT adversary* $\mathcal{A}$ *we have* $Adv^{pay}_{\text{PQ-RDHED},\mathcal{A}}(\lambda) \leq 2 \cdot Adv^{PRF}_{PRF,\mathcal{B}}(\lambda)$. *Under the standard key model with* $hk = xk$ *(Remark 6), payload privacy reduces to semantic security with* $Adv^{pay}_{\text{PQ-RDHED},\mathcal{A}}(\lambda) \leq Adv^{sem}_{\text{PQ-RDHED},\mathcal{A}'}(\lambda)$.

Proof **Case 1 (Extended Key Model,** $hk \neq xk$**)** Consider the experiment $\mathsf{Expt}^{\mathsf{pay}}_{\mathcal{A},b}(\lambda)$ where $\mathcal{A}$ receives (ek, hk) but not xk.

Game G_0 The real payload privacy experiment. The challenger computes $\tilde{d} = d_b \oplus \mathsf{PRF}(K_{\mathsf{mask}}, 0\text{x}00)[0 : |d_b|]$ where $K_{\mathsf{mask}} = \mathsf{H}(K_h \| 0\text{x}03)$.

Game G_1 Replace $\mathsf{PRF}(K_{\mathsf{mask}}, 0\text{x}00)[0 : |d_b|]$ with $r \leftarrow \{0,1\}^{|d_b|}$ sampled uniformly.

Claim $|\Pr[G_0 = 1] - \Pr[G_1 = 1]| \leq \mathsf{Adv}^{\mathsf{PRF}}_{\mathsf{PRF},\mathcal{B}}(\lambda)$.

Proof Construct $\mathcal{B}$ with oracle $\mathcal{O}$ (either $\mathsf{PRF}(K_{\mathsf{mask}}, \cdot)$ or random). $\mathcal{B}$ simulates the experiment, querying $\mathcal{O}(0\text{x}00)$ to obtain the mask. Distinguishing G_0 from G_1 implies distinguishing PRF from random. In G_1, $\tilde{d} = d_b \oplus r$ where r is uniform random. For any d_0, d_1 with $|d_0| = |d_1|$: $\tilde{d_0} = d_0 \oplus r \overset{d}{=} r \overset{d}{=} d_1 \oplus r = \tilde{d_1}$, where $\overset{d}{=}$ denotes identical distribution. Thus $\Pr[G_1 = 1 \mid b = 0] = \Pr[G_1 = 1 \mid b = 1]$, implying $\mathsf{Adv}^{\mathsf{pay}}_{G_1} = 0$. For Case 1, the advantage bound will be:

$$\begin{aligned}\mathsf{Adv}^{\mathsf{pay}}_{\text{PQ-RDHED},\mathcal{A}}(\lambda) &= |\Pr[G_0 = 1 \mid b = 0] - \Pr[G_0 = 1 \mid b = 1]| \\ &\leq |\Pr[G_0 = 1] - \Pr[G_1 = 1]| + |\Pr[G_1 = 1 \mid b = 0] - \Pr[G_1 = 1 \mid b = 1]| \\ &\quad + |\Pr[G_1 = 1] - \Pr[G_0 = 1]| \leq 2 \cdot \mathsf{Adv}^{\mathsf{PRF}}_{\mathsf{PRF},\mathcal{B}}(\lambda) + 0.\end{aligned}$$

Case 2 (Standard Key Model, $hk = xk$**)** ***When*** $\mathcal{A}$ ***possesses*** $hk = K_h = K_x = xk$***, it can derive*** K_{mask} ***and unmask*** $\tilde{d}$ ***to recover*** d_b***. In this setting, payload privacy is implied by semantic security: if*** $\mathcal{A}$ ***could distinguish payloads, construct*** $\mathcal{A}'$ ***that embeds*** d_0, d_1 ***into encryptions of distinct messages and uses*** $\mathcal{A}$***'s output to distinguish ciphertexts. Thus*** $\mathsf{Adv}^{\mathsf{pay}} \leq \mathsf{Adv}^{\mathsf{sem}}$. □

Theorem 5 *The proposed* PQ-RDHED *scheme (Definition 16) satisfies the property of separability (Definition 11).*

Proof We verify both conditions of Definition 11.

Condition (I) - Extraction-First We show $\mathsf{Extract}(xk, \cdot)$ is computable without dk. Examining Extract in Definition 16:

- Step 1 derives $(K_{\mathsf{pos}}, K_{\mathsf{mask}})$ from $xk = K_x$ via H; no dependence on $dk = sk$.
- Steps 2–4 perform MSB analysis and Huffman decoding on encrypted blocks $\{\hat{E}_k\}$; operations are combinatorial on ciphertext.
- Steps 5–8 parse, verify, and unmask the payload using K_{mask}; verification uses $\Pi_{\mathsf{LA}}.\mathsf{VerifyShare}$ which requires only π_{share}.
- Steps 9–11 reconstruct c from aux metadata; no decryption invoked.

Thus $\mathsf{Extract} : \{0,1\}^\lambda \times \hat{\mathcal{C}} \to \mathcal{C} \times \mathcal{D}$ depends only on $(xk, \hat{c})$.

Condition (II) - Decryption-First Define $\mathsf{Dec}' : \{0,1\}^* \times \hat{\mathcal{C}} \to \hat{\mathcal{M}}$ as follows: Parse $\hat{c} = (c_{\mathsf{kem}}, \{\hat{E}_k\}_{k=1}^{\ell}, \mathsf{tag})$. Compute $K \leftarrow \mathsf{ML\text{-}KEM.Decaps}(sk, c_{\mathsf{kem}})$. Derive $K_e = \mathsf{H}(K\|0\mathrm{x}00), K_\pi = \mathsf{H}(K\|0\mathrm{x}01)$. Compute $\sigma^{-1} = \mathsf{FisherYates}(\ell, K_\pi)^{-1}$. For each $k \in [\ell]$: set $\hat{E}'_k = \hat{E}_{\sigma(k)}$; compute $\mu_k = \mathsf{PRF}(K_e, k)[0:8]$; for each $u \in [n_b^2]$: $\hat{p}_{k,u} = \hat{E}'_k[u] \oplus \mu_k$. Output $\hat{m} = \{\hat{p}_{k,u}\}_{k\in[\ell],u\in[n_b^2]}$. The output $\hat{m}$ encodes: (i) the original image m in the $(8-\beta)$ LSB positions of each pixel; (ii) the embedded stream (Huffman-encoded MSBs interleaved with payload bits) in the β MSB positions of pixels in embeddable blocks. Formally, for embeddable block k and pixel u:

$$\hat{p}_{k,u} = \underbrace{(0)^{p_{k,u}} \| 1 \| \overline{\mathsf{code}_k(\mathsf{msb}_{k,u})}}_{\beta \text{ MSBs: encoded data}} \| \underbrace{(p_{k,u} \mod 2^{8-\beta})}_{(8-\beta) \text{ LSBs: original}}.$$

Recovery of (m, d) from $\hat{m}$ requires only the location map L (identifiable via encoding structure) and Huffman decoding tables (reconstructible from α_k counts), both extractable from $\hat{m}$ without xk. □

Theorem 6 *The proposed* PQ-RDHED *scheme (Definition 16) satisfies property of compactness (Definition 12) with overhead $p(\lambda, |d|) = \lambda$. Specifically, for all valid (c, d) we will have $|Embed(hk, c, d)| = |c| + \lambda$.*

Proof of Sketch Let $c = (c_{\mathsf{kem}}, \{E'_k\}_{k=1}^{\ell})$ and $\hat{c} = (c_{\mathsf{kem}}, \{\hat{E}_k\}_{k=1}^{\ell}, \mathsf{tag})$. To consider component sizes, $|c_{\mathsf{kem}}|$ is fixed by ML-KEM parameters; identical in c and $\hat{c}$, $|\{E'_k\}_{k=1}^{\ell}|$ is $\ell \cdot n_b^2 \cdot 8$ bits (each pixel is 8 bits), $|\{\hat{E}_k\}_{k=1}^{\ell}|$ is $\ell \cdot n_b^2 \cdot 8$ bits. Embedding modifies only the β-MSB positions within existing pixels; no expansion occurs since Huffman encoding plus payload occupies exactly $\beta \cdot n_b^2$ bits per embeddable block (the same space as the original MSBs); and $|\mathsf{tag}|$ is λ bits (output length of H). To compute the total sizes here, $|c| = |c_{\mathsf{kem}}| + \ell \cdot n_b^2 \cdot 8$, $|\hat{c}| = |c_{\mathsf{kem}}| + \ell \cdot n_b^2 \cdot 8 + \lambda = |c| + \lambda$. Thus $|\hat{c}| \le |c| + p(\lambda, |d|)$ with $p(\lambda, |d|) = \lambda$, satisfying Definition 12. □

Theorem 7 PQ-RDHED *satisfies embedding robustness (Definition 13) in the QROM. For any QPT adversary $\mathcal{A}$ making at most q_H quantum queries to H we have* $Adv^{robust}_{\mathsf{PQ\text{-}RDHED},\mathcal{A}}(\lambda) \le Adv^{CR}_{H,\mathcal{B}_1}(\lambda) + Adv^{bind}_{\Pi_{LA},\mathcal{B}_2}(\lambda) + \frac{q_H^2}{2^\lambda}$.

Proof Consider $\mathcal{A}$ outputting $(c, d, \hat{c}^*)$ such that $\hat{c}^* \neq \mathsf{Embed}(hk, c, d)$ but $\mathsf{Extract}(xk, \hat{c}^*) = (c, d)$. Let $\hat{c} := \mathsf{Embed}(hk, c, d) = (c_{\mathsf{kem}}, \{\hat{E}_k\}, \mathsf{tag})$ be the honest embedding, and parse $\hat{c}^* = (c^*_{\mathsf{kem}}, \{\hat{E}^*_k\}, \mathsf{tag}^*)$.

Case 1 $c^*_{\mathsf{kem}} \neq c_{\mathsf{kem}}$. The extraction algorithm verifies $\mathsf{tag}^* = \mathsf{H}(K_x \| c^*_{\mathsf{kem}} \| \mathsf{H}(\mathsf{stream}'))$. For extraction to yield $c = (c_{\mathsf{kem}}, \cdot)$ from $\hat{c}^* = (c^*_{\mathsf{kem}}, \cdot, \cdot)$ with $c^*_{\mathsf{kem}} \neq c_{\mathsf{kem}}$, we require the recovered ciphertext to contain c_{kem}. However, Extract outputs $c' = (c^*_{\mathsf{kem}}, \{E'_k\})$, preserving c^*_{kem}. Thus $c' \neq c$, contradiction.

Case 2 $c^*_{\mathsf{kem}} = c_{\mathsf{kem}}$ and $\{\hat{E}^*_k\} \neq \{\hat{E}_k\}$. Let stream and stream' denote the embedded streams for $\hat{c}$ and $\hat{c}^*$ respectively. Since $\mathsf{Extract}(xk, \hat{c}^*) = (c, d)$, then the payload satisfies $d' = d$ (where $d' = \tilde{d}' \oplus \mathsf{PRF}(K_{\mathsf{mask}}, 0\mathrm{x}00)[0 : |d'|]$), the auxiliary data

Table 12 Which mechanism prevents each modification, and where it is discharged in the proof of Theorem 7

Component modified by $\mathcal{A}$	Protecting mechanism	Proof case
Payload d	Π_{LA} commitment to $\mathsf{H}(\mathsf{aux}\|\|d)$, then tag	2b-i, 2b-ii
Location map $L \subseteq \mathsf{aux}$	Π_{LA} commitment to $\mathsf{H}(\mathsf{aux}\|\|d)$, then tag	2b-i, 2b-ii
MSB tables $(\alpha_k, \mathsf{msb}_k^{(j)}) \subseteq \mathsf{aux}$	Π_{LA} commitment to $\mathsf{H}(\mathsf{aux}\|\|d)$, then tag	2b-i, 2b-ii
Embeddable blocks $\hat{E}_k$, $k \in \mathcal{E}$	tag binds $\mathsf{H}(\mathsf{stream})$	2a, 3
Non-embeddable blocks $\hat{E}_k$, $k \notin \mathcal{E}$	returned verbatim, so $c' \neq c$	1
KEM ciphertext c_{kem}	tag binds c_{kem}; echoed into c'	1
Authentication metadata tag	H keyed by the secret $hk = xk$ (QROM)	3

satisfies $\mathsf{aux}' \to c$ (reconstruction yields original ciphertext), and the VSS satisfies $\Pi_{\mathsf{LA}}.\mathsf{Reconstruct}(\cdot, \pi'_{\mathsf{share}}) = \mathsf{H}(\mathsf{aux}'\|d)$.

Subcase 2a $\mathsf{stream}' = \mathsf{stream}$. Since embedding is deterministic given (hk, c, stream), and $\mathsf{stream}' = \mathsf{stream}$ implies identical embedding bits, we have $\{\hat{E}_k^*\} = \{\hat{E}_k\}$, contradicting the assumption.

Subcase 2b $\mathsf{stream}' \neq \mathsf{stream}$ with $(\mathsf{aux}', d', \pi'_{\mathsf{share}}) \neq (\mathsf{aux}, d, \pi_{\mathsf{share}})$.

Subcase 2b-i If $(\mathsf{aux}', d') \neq (\mathsf{aux}, d)$ but $\mathsf{H}(\mathsf{aux}'\|d') = \mathsf{H}(\mathsf{aux}\|d)$ (required for VSS verification against the same commitment), then $(\mathsf{aux}'\|d', \mathsf{aux}\|d)$ constitutes a collision in H. Construct $\mathcal{B}_1$: run $\mathcal{A}$ to obtain $(c, d, \hat{c}^*)$; compute honest embedding $\hat{c}$; extract stream' from $\hat{c}^*$; if $\mathsf{aux}'\|d' \neq \mathsf{aux}\|d$ and both hash to the same value, output the collision pair. Thus: $\Pr[\text{Subcase 2b-i}] \leq \mathsf{Adv}^{\mathsf{CR}}_{\mathsf{H},\mathcal{B}_1}(\lambda)$.

Subcase 2b-ii If $\pi'_{\mathsf{share}} \neq \pi_{\mathsf{share}}$ but both verify and reconstruct to $\mathsf{H}(\mathsf{aux}\|d)$, this violates the binding property of Π_{LA}. Construct $\mathcal{B}_2$: run $\mathcal{A}$; extract both πs; if distinct πs reconstruct to the same value, output the binding break. Thus: $\Pr[\text{Subcase 2b-ii}] \leq \mathsf{Adv}^{\mathsf{bind}}_{\Pi_{\mathsf{LA}},\mathcal{B}_2}(\lambda)$.

Case 3 $\mathsf{tag}^* \neq \mathsf{tag}$ with $(c^*_{\mathsf{kem}}, \{\hat{E}_k^*\}) = (c_{\mathsf{kem}}, \{\hat{E}_k\})$.Then $\hat{c}^* = (c_{\mathsf{kem}}, \{\hat{E}_k\}, \mathsf{tag}^*)$ with $\mathsf{tag}^* \neq \mathsf{tag}$. The extraction verifies $\mathsf{tag}^* = \mathsf{H}(K_x\|c_{\mathsf{kem}}\|\mathsf{H}(\mathsf{stream}'))$. Since embedded blocks are identical, $\mathsf{stream}' = \mathsf{stream}$, hence the correct tag is $\mathsf{tag} = \mathsf{H}(K_x\|c_{\mathsf{kem}}\|\mathsf{H}(\mathsf{stream}))$. If $\mathsf{tag}^* \neq \mathsf{tag}$, verification fails and extraction outputs $\perp \neq (c, d)$, contradiction. Finally, for the total advantage we will have:

$$\begin{aligned}\mathsf{Adv}^{\mathsf{robust}}_{\mathsf{PQ\text{-}RDHED},\mathcal{A}}(\lambda) &\leq \Pr[\text{Case 2b-i}] + \Pr[\text{Case 2b-ii}] + \Pr[\text{QROM collision}] \\ &\leq \mathsf{Adv}^{\mathsf{CR}}_{\mathsf{H},\mathcal{B}_1}(\lambda) + \mathsf{Adv}^{\mathsf{bind}}_{\Pi_{\mathsf{LA}},\mathcal{B}_2}(\lambda) + \frac{q_H^2}{2^\lambda}.\end{aligned}$$

□

Remark 7 (Coverage of the Robustness Argument). Since robustness is a central claim, we make explicit which mechanism protects each modifiable component of $\hat{c}$ and which case of the proof of Theorem 7 discharges it; Table 12 summarises the correspondence. The key point is that the two guards are nested. The VSS commitment binds a *single* digest $\mathsf{H}(\mathsf{aux}\|d)$

that covers the payload and the whole of aux, that is, the location map L, the count $|\mathcal{E}|$, and the per-block $(\alpha_k, \{\mathsf{msb}_k^{(j)}\})$ tables, so any change to the embedded MSB metadata or to the location map changes the digest, and passing $\Pi_{\mathsf{LA}}.\mathsf{VerifyShare}$ afterwards requires either a collision in H (Subcase 2b-i) or two distinct proofs opening the same commitment, which breaks binding (Subcase 2b-ii). The commitment, however, does not itself cover the ciphertext blocks or the KEM ciphertext, because those are not inputs to $\Pi_{\mathsf{LA}}.\mathsf{Share}$; they are covered instead by $\mathsf{tag} = \mathsf{H}(hk \| c_{\mathsf{kem}} \| \mathsf{H}(\mathsf{stream}))$, whose preimage contains c_{kem} explicitly and every embeddable block implicitly through stream. A non-embeddable block carries no payload and is reproduced verbatim by Extract, so altering it changes the recovered c' and the adversary loses on the condition $c' = c$ rather than on a cryptographic guard (Case 1). Finally, tag itself is authenticated by the secret hk: an adversary that has not queried H on $hk \| c_{\mathsf{kem}} \| \cdot$ can produce a matching tag only by guessing, which in the QROM is bounded by $q_H^2/2^\lambda$ through the standard collision bound (Case 3).

Under the quantum hardness of $\mathsf{MLWE}_{n,k,q,\chi}$, post-quantum collision resistance of H in the QROM, and quantum-secure instantiations of PRF (via SHAKE256) and Π_{LA} (hash-based VSS), all security properties established in Theorems 1–7 hold against QPT adversaries. Concretely, for any QPT adversary $\mathcal{A}$ and security parameter λ:

$$\mathsf{Adv}_{\mathcal{A}}^{\mathsf{sem}}(\lambda), \mathsf{Adv}_{\mathcal{A}}^{\mathsf{pay}}(\lambda), \mathsf{Adv}_{\mathcal{A}}^{\mathsf{unlink}}(\lambda), \mathsf{Adv}_{\mathcal{A}}^{\mathsf{robust}}(\lambda) = \mathsf{negl}(\lambda).$$

The reductions in Theorems 3, 4, and 7 are polynomial-time and preserve quantum query complexity. By [32], $\mathsf{Adv}_{\mathsf{ML\text{-}KEM}}^{\mathsf{IND\text{-}CCA}}(\lambda) = \mathsf{negl}(\lambda)$ under $\mathsf{MLWE}_{n,k,q,\chi}$ in the QROM. By [33], SHAKE256 instantiates a quantum-secure PRF with $\mathsf{Adv}_{\mathsf{PRF}}^{\mathsf{PRF}}(\lambda) = \mathsf{negl}(\lambda)$. By [34], SHA3-256 provides $\mathsf{Adv}_{\mathsf{H}}^{\mathsf{CR}}(\lambda) = \mathsf{negl}(\lambda)$ against quantum adversaries. The VSS scheme Π_{LA} inherits post-quantum binding from its hash-based commitments [35], yielding $\mathsf{Adv}_{\Pi_{\mathsf{LA}}}^{\mathsf{bind}}(\lambda) = \mathsf{negl}(\lambda)$. Substituting into the advantage bounds:

$$\mathsf{Adv}_{\mathcal{A}}^{\mathsf{sem}}(\lambda) \le 2 \cdot \mathsf{negl}(\lambda) + 2\ell \cdot \mathsf{negl}(\lambda) = \mathsf{negl}(\lambda),$$
$$\mathsf{Adv}_{\mathcal{A}}^{\mathsf{robust}}(\lambda) \le \mathsf{negl}(\lambda) + \mathsf{negl}(\lambda) + 2^{-\Omega(\lambda)} = \mathsf{negl}(\lambda).$$

The *security reductions* of Theorems 3, 4 and 7 are unconditional as they exhibit explicit QPT algorithms $\mathcal{B}_i$ and hold whatever the true hardness of the underlying problems turns out to be. The *asymptotic* claim, that every advantage is $\mathsf{negl}(\lambda)$, is obtained by substituting the assumed negligibility of the building blocks and is therefore exactly as strong as those assumptions, no stronger. The *concrete* security figures 2^{-128} and 2^{-127} quoted for the recommended parameter set are estimates that inherit the current best cryptanalysis of MLWE at the ML-KEM-768 parameters and the QROM collision bound $q_H^2/2^\lambda$; they are not proved lower bounds and will move if lattice cryptanalysis [39] improves. Finally, none of these three statements is a claim about *implementation* security: the QROM is an idealised model with no instantiation guarantee, and the proofs say nothing about timing or cache side channels, fault injection, key-management failures, or the randomness quality of the sampler used for ML-KEM.KeyGen and $\Pi_{\mathsf{LA}}.\mathsf{Share}$. A deployment of PQ-RDHEI therefore requires constant-time implementations of the building blocks and the usual side-channel countermeasures, which are orthogonal to the guarantees established here.

7 Conclusion

This work presents the first formal cryptographic treatment of Reversible Data Hiding in Encrypted Images. We have defined RDHEI as a cryptographic primitive with well-defined syntax, accurate correctness conditions and security traits such as semantic security, payload privacy, separability, compactness, and robustness. Following this formalization, PQ-RDHEI was developed, which is post-quantum secure under the MLWE assumption in the quantum random oracle model. The construction relies on ML-KEM for key encapsulation, pseudorandom functions realized through SHAKE256, the SHA3-256 hash function, and hash-based verifiable secret sharing. Our experimental results show that PQ-RDHEI not only allows for greater embedding capacity than previous methods but also provides formal security guarantees that were not available in the literature before.

As future work, one direction can be focusing on the efficiency of RDHEI by a more formal approach; e.g., apart from compactness, we can say an RDHEI scheme has α-*capacity efficiency* for $\alpha \in (0,1]$ if $\kappa(n) \geq \alpha \cdot n$ for all $n \in \mathbb{N}$, and this definition can have further development. Another future direction can be improving our PQ-RDHEI to achieve the strong variant of compactness (Definition 12); strong compactness can be achieved in PQ-RDHEI by omitting tag at the cost of robustness guarantees, but without omitting that it is not possible due to the λ-bit authentication tag. Considering the limitation of RDHEI methods in gray-scale, extending the formalization to color images presents additional research opportunities. Moreover, we aim to eliminate the secret sharing phase in RDHEI totally via silent-setup threshold cryptosystems [40] which boosts general efficiency; however, current silent-setup constructions rely on witness encryption, which remains impractical and quantum-vulnerable [41–43], so designing a post-quantum silent-setup scheme supporting homomorphic operations to get used in PQ-RDHEI is another open problem.

The single limitation of this work, and the one we regard as the most promising direction for follow-up research, is that the message space is fixed to grayscale images, $\mathcal{M} = \{0, \ldots, 255\}^{W \times H}$. This restriction is deliberate on the cryptographic side and genuine on the multimedia side, and the two aspects should be separated. Cryptographically it costs nothing: the encryption layer is a KEM-DEM construction over an arbitrary byte string, so applying PQ-RDHEI independently to the R, G and B planes of a colour image under domain-separated keys $K_e^{(\chi)} = \mathsf{H}(K \| 0\text{x}00 \| \chi)$ for $\chi \in \{R, G, B\}$ yields a scheme for which Theorems 1–7 carry over verbatim by a standard hybrid argument over the three planes, at a cost of two additional terms in each advantage bound. Every security and correctness statement of this paper is therefore already colour-ready, which is why we present the formalization in the simpler single-plane setting: the grayscale message space isolates the cryptographic content of the definitions from the channel-representation questions that colour introduces, and no definitional machinery is lost. What such a plane-wise extension does *not* do is exploit colour, and this is the real limitation. A per-plane treatment ignores the strong inter-channel correlation of natural colour images, correlation coefficients between the R, G and B planes typically exceed 0.9, so it leaves capacity on the table relative to a scheme that predicts one plane from another, and it does not address chroma-subsampled or non-RGB representations such as $\mathrm{YC_bC_r}$, in which the planes have different resolutions and the block partition of Definition 16 no longer aligns across channels. Designing a capacity function κ and an embeddability criterion that operate jointly on the three planes, and proving that the resulting cross-channel prediction does not leak plaintext structure into the marked ciphertext, is a genuinely open problem and, in our view, the natural next step for this line of work. We

note finally that the grayscale restriction is least binding in exactly the application domains that motivate the construction, since computed-tomography, magnetic-resonance, ultrasound and radiographic images, evaluated in Section 5.2, are single-channel by acquisition.

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Data Availability All implementation details are available from the author on request. The data used in this study are standard benchmark images obtained from the publicly available USC-SIPI Image Database available at http://sipi.usc.edu/database, and medical images from the openly accessible MedPix collection served by the Open-i service of the U.S. National Library of Medicine [38], available at https://openi.nlm.nih.gov. No patient-identifiable information is contained in any of the images used.

Declarations

Ethics approval and consent to participate Not applicable

Consent for publication The author has consent for the publication.

Competing interests The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

1. Tian J (2003) Reversible data embedding using a difference expansion. IEEE Trans Circuits Syst Video Technol 13(8):890–896
2. Ni Z, Shi Y-Q, Ansari N, Su W (2006) Reversible data hiding. IEEE Trans Circuits Syst Video Technol 16(3):354–362
3. Thodi DM, Rodríguez JJ (2007) Expansion embedding techniques for reversible watermarking. IEEE Trans Image Process 16(3):721–730
4. Shi Y-Q, Li X, Zhang X, Wu H, Ma B (2016) Reversible data hiding: Advances in the past two decades. IEEE Access 4:3210–3237
5. Dragoi I-C, Coltuc D (2014) Local-prediction-based difference expansion reversible watermarking. IEEE Trans Image Process 23(4):1779–1790
6. Beggari AS, Wali A, Khaldi A, Kafi MR, Sahu AK (2025) Secure and imperceptible medical image watermarking via multiscale QR embedding and attention-based optimization. Eng Sci Technol Intern J 73:102250. https://doi.org/10.1016/j.jestch.2025.102250
7. Sachnev V, Kim HJ, Nam J, Suresh S, Shi YQ (2009) Reversible watermarking algorithm using sorting and prediction. IEEE Trans Circuits Syst Video Technol 19(7):989–999
8. Zhang X, Ren Y, Shen L, Qian Z, Feng G (2014) Compressing encrypted images with auxiliary information. IEEE Trans Multimedia 16(5):1327–1336
9. Ke Y, Zhang M-Q, Liu J, Su T-T, Yang X-Y (2020) Fully homomorphic encryption encapsulated difference expansion for reversible data hiding in encrypted domain. IEEE Trans Circuits Syst Video Technol 30(8):2353–2365

10. Ch R, Shaik J, Srikavya R et al (2025) A novel fiestal structured chromatic series-based data security approach. Discover Internet of Things 5:106. https://doi.org/10.1007/s43926-025-00162-0
11. Puech W, Chaumont M, Strauss O (2008) A reversible data hiding method for encrypted images. In: Security, forensics, steganography, and watermarking of multimedia contents X, vol 6819. SPIE, pp 68191
12. Zhang X (2011) Reversible data hiding in encrypted image. IEEE Signal Process Lett 18(4):255–258
13. Liao X, Shu C (2015) Reversible data hiding in encrypted images based on absolute mean difference of multiple neighboring pixels. J Vis Commun Image Represent 28:21–27
14. Zhang X (2012) Separable reversible data hiding in encrypted image. IEEE Trans Inf Forensics Secur 7(2):826–832
15. Ma K, Zhang W, Zhao X, Yu N, Li F (2013) Reversible data hiding in encrypted images by reserving room before encryption. IEEE Trans Inf Forensics Secur 8(3):553–562
16. Puteaux P, Puech W (2018) An efficient MSB prediction-based method for high-capacity reversible data hiding in encrypted images. IEEE Trans Inf Forensics Secur 13(7):1670–1681
17. Yin Z, Xiang Y, Zhang X (2020) Reversible data hiding in encrypted images based on multi-MSB prediction and Huffman coding. IEEE Trans Multimedia 22(4):874–884
18. Mohammadi A, Nakhkash M, Akhaee MA (2020) A high-capacity reversible data hiding in encrypted images employing local difference predictor. IEEE Trans Circ Syst Video Technol 30(8):2366–2376
19. Wu X, Weng J, Yan W (2018) Adopting secret sharing for reversible data hiding in encrypted images. Signal Process 143:269–281
20. Chen Y-C, Hung T-H, Hsieh S-H, Shiu C-W (2019) A new reversible data hiding in encrypted image based on multi-secret sharing and lightweight cryptographic algorithms. IEEE Trans Inf Forensics Secur 14(12):3332–3343
21. Qin C, Jiang C, Mo Q, Yao H, Chang C-C (2022) Reversible data hiding in encrypted image via secret sharing based on GF(p) and GF(2^8). IEEE Trans Circuits Syst Video Technol 32(4):1928–1941
22. Chen B, Lu W, Huang J, Weng J, Zhou Y (2022) Secret sharing based reversible data hiding in encrypted images with multiple data-hiders. IEEE Trans Dependable Secure Comput 19(2):978–991
23. Hua Z, Wang Y, Yi S, Zhou Y, Jia X (2022) Reversible data hiding in encrypted images using cipher-feedback secret sharing. IEEE Trans Circuits Syst Video Technol 32(8):4968–4982
24. Saeidi Z, Mirzakuchaki S (2025) Reversible data hiding in encrypted images using LWE-based secret sharing. Sci Rep 15(1):44819. https://doi.org/10.1038/s41598-025-28912-8
25. Regev O (2009) On lattices, learning with errors, random linear codes, and cryptography. J ACM 56(6)
26. Rogaway P (2016) Practice-oriented provable security and the social construction of cryptography. IEEE Secur Priv 14(6):10–17. https://doi.org/10.1109/MSP.2016.122
27. Diaz Santiago S, Gallegos Garcia G, Rodriguez Henriquez LM, Cruz Cortes N (2016) A brief introduction to provable security. IEEE Lat Am Trans 14(5):2323–2332. https://doi.org/10.1109/TLA.2016.7530429
28. Bellare M (1998) Practice-oriented provable security. In: Information Security Workshop (ISW), vol 1396. ILNCS, pp 1–15
29. Katz J, Lindell Y (2020) Introduction to Modern Cryptography, 3rd edn. CRC Press, Boca Raton, FL, USA
30. Shor PW (1994) Algorithms for quantum computation: Discrete logarithms and factoring. In: Proceedings 35th annual symposium on foundations of computer science. IEEE, pp 124–134
31. Mahdavi M, Abapour N, Ahmadian Z (2024) Trustworthy Approaches to RSA: Efficient Exploitation Strategies Based on Common Modulus. Cryptology ePrint , Paper 2024/1903
32. National Institute of Standards and Technology (2024) Module-lattice-based key-encapsulation mechanism standard. Technical Report Federal Information Processing Standards Publication (FIPS) 203, U.S. Department of Commerce, Gaithersburg, MD, USA. https://doi.org/10.6028/NIST.FIPS.203
33. National Institute of Standards and Technology (2015) SHA-3 Standard: Permutation-Based Hash and Extendable-Output Functions. Technical Report Federal Information Processing Standards Publication (FIPS) 202, U.S. Department of Commerce. https://doi.org/10.6028/NIST.FIPS.202
34. Bertoni G, Daemen J, Peeters M, Van Assche G (2013) Keccak. In: Advances in cryptology – EUROCRYPT 2013. Lecture notes in computer science, vol 7881, pp 313–314. https://doi.org/10.1007/978-3-642-38348-9_19
35. Baghery K (2025) Π: A unified framework for computational verifiable secret sharing. In: Jager T, Pan J (eds) Public-Key Cryptography – PKC 2025. LNCS, vol 15677. https://doi.org/10.1007/978-3-031-91829-2_4
36. Durstenfeld R (1964) Algorithm 235: Random permutation. Commun ACM 7(7):420. https://doi.org/10.1145/364520.364540

37. Grover LK (1996) A fast quantum mechanical algorithm for database search. In: Proceedings of the twenty-eighth annual ACM symposium on theory of computing. STOC '96. Association for Computing Machinery, New York, NY, USA, pp 212–219. https://doi.org/10.1145/237814.237866
38. Demner-Fushman D, Antani S, Simpson M, Thoma GR (2012) Design and development of a multimodal biomedical information retrieval system. J Comput Sci Eng 6(2):168–177. https://doi.org/10.5626/JCSE.2012.6.2.168
39. Abapour N (2026) Sharper and closed-form attacks on SIS when the modulus is small. IACR Commun Cryptol 3(2). https://doi.org/10.62056/abe0zogxq
40. Garg S, Kolonelos D, Policharla G-V, Wang M (2024) Threshold encryption with silent setup. In: Advances in cryptology – CRYPTO 2024, pp 352–386. https://doi.org/10.1007/978-3-031-68394-7_12
41. Waters B, Wu DJ (2025) Silent threshold cryptography from pairings: Expressive policies in the plain model. Cryptology ePrint Archive, Paper 2025/1547. https://eprint.iacr.org/2025/1547
42. Hall-Andersen M, Simkin M, Wagner B (2025) Silent threshold encryption with one-shot adaptive security. Cryptology ePrint Archive, Paper 2025/1384. https://eprint.iacr.org/2025/1384
43. Bormet J et al (2025) BEAST-MEV: Batched threshold encryption with silent setup for MEV prevention. Cryptology ePrint Archive, Paper 2025/1419

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